NASH BARGAINING PREFERENCE OPTIMIZATION

Anonymous authors
Paper under double-blind review

ABSTRACT

Aligning large language models (LLMs) to multiple objectives requires both representing preferences within each objective and selecting a compromise across objectives. Human comparisons need not admit a global scalar reward: annotators may weigh latent attributes differently across contexts and response pairs, so aggregate judgments can be cyclic. Most existing methods nevertheless represent each objective through scalar scores over individual responses and cannot capture such cycles. Nash Learning from Human Feedback (NLHF) instead treats prompt-conditioned pairwise preferences directly as game payoffs. A recent multi-objective NLHF approach retains this generality but uses max-min aggregation. However, max-min can select Pareto-dominated outcomes and change under objective rescaling, so the resulting compromise lacks basic efficiency and scale robustness. We introduce Nash Bargaining Preference Optimization (NBPO), which separates these decisions. An entropy-regularized preference game yields one value per objective, and Nash bargaining balances improvements over a reference fallback. The reference becomes a meaningful status quo, while the trade-off follows proportional gains rather than prespecified weights. When joint improvement is feasible, NBPO strictly improves every objective over the fallback, selects a Pareto-optimal and unique game-value vector, is proportionally fair, and leaves the set of optimal policies unchanged under independent positive affine transformations of objective utilities. A closed-form regularized opponent avoids training an additional opponent model, while partition-free pairwise log-ratio regression removes policy normalizers and learns from binary comparisons without fitting scalar rewards. The resulting constrained entropy-proximal solver starts directly from the reference, recovers inverse-surplus weights, and admits last-iterate convergence with explicit sampling and regression errors.

1 INTRODUCTION

Preference-based post-training is central to aligning Large Language Models (LLMs) with human judgments (Christiano et al., 2017; Ouyang et al., 2022). Standard Reinforcement Learning from Human Feedback (RLHF) learns a scalar reward from pairwise comparisons, while Direct Preference Optimization removes the separately trained reward model but retains an implicit scalar reward and a Bradley–Terry (BT) representation of preferences (Rafailov et al., 2023; Bradley & Terry, 1952). Both approaches require comparisons to be explained by a single scalar score assigned to each response. This assumption is restrictive for multi-objective alignment, where one deployed assistant must satisfy objectives such as helpfulness, harmlessness, correctness, and conciseness.

Even within one named objective, human judgments may depend on several latent attributes whose relative importance changes across annotators, prompts, and response pairs. Consider three answers to the same prompt under a helpfulness criterion. Response A is direct but omits important qualifications, response B is comprehensive but verbose, and response C is clear and well qualified but less complete. Aggregate comparisons may prefer A to B for relevance, B to C for completeness, and C to A for clarity and reliability. No scalar reward can represent all three preferences simultaneously. For example, fitting $A \succ B$ and $B \succ C$ requires $r(A) > r(B) > r(C)$, which contradicts the observed preference $C \succ A$. A BT reward model must therefore misrepresent at least one pairwise comparison in order to express the cyclic feedback as a transitive scalar ranking. Optimizing this projection aligns the policy to the fitted ordering rather than the original comparison structure, so a scalar-optimal response can remain vulnerable to another response under the observed preferences.

Nash Learning from Human Feedback (NLHF) avoids this misspecification by treating prompt-conditioned comparisons directly as payoffs in a zero-sum preference game, while INPO, SPPO, and related methods provide scalable equilibrium-seeking updates (Munos et al., 2024; Zhang et al., 2025; Wu et al., 2025; Calandriello et al., 2024; Rosset et al., 2024; Zhou et al., 2025). Extending this view to multiple objectives introduces a separate question: after representing each objective faithfully, which compromise should one deployed policy realize? Existing scalar multi-objective methods require an external trade-off specification, such as weights, priorities, or constraints (Zhou et al., 2024; Ramé et al., 2023; Zhong et al., 2024b; Chakraborty et al., 2024; Son et al., 2026; Agnihotri et al., 2026), whereas recent game-theoretic extensions use Blackwell target sets or max-min aggregation (Blackwell, 1956; Bhatia et al., 2020; Zhang et al., 2026). Fixed-reference methods such as MOPO are computationally convenient but can discard pairwise structure that is invisible from the reference (Agnihotri et al., 2026; Zhang et al., 2026).

We study the complementary setting in which a fixed objective panel must be combined into one policy without prespecified priorities. Here, representing preferences within each objective and aggregating values across objectives are distinct decisions. The first calls for a game value that retains the full pairwise structure; the second calls for an explicit compromise rule. A max-min game couples the two by imposing a weakest-objective rule while evaluating the policy. It can retain Pareto-dominated tied solutions and can change under independent rescaling of objective utilities even when the underlying preferences are unchanged.

We introduce Nash Bargaining Preference Optimization (NBPO), which separates these roles. For each objective, an adaptive opponent searches for responses that expose weaknesses of the current policy, while entropy regularization keeps the opponent near the deployed reference. This produces a game value based on the full pairwise comparison matrix rather than a scalar response score or only a fixed-reference margin. The reference also defines the bargaining fallback, and NBPO selects a policy by balancing positive game-value improvements over it. Nash bargaining is natural because it evaluates each objective relative to a meaningful status quo, gives more influence to objectives with smaller gains, and requires no externally chosen objective weights.

When joint improvement over the reference is feasible, every NBPO optimizer improves every objective strictly beyond the fallback and is Pareto optimal in game-value space. Different policies may realize this outcome, but the selected game-value vector is unique. The policy selection is unchanged by independent positive affine transformations of the objective utility scales. Utilitarian and max-min aggregation are useful alternatives, but neither guarantees this full combination of fallback-relative protection, efficiency, uniqueness, and scale covariance without additional restrictions or tie-breaking.

The optimizer preserves the same separation. We impose positive game-value improvement as a constraint and solve entropy-proximal bargaining subproblems directly from the reference boundary, rather than optimizing a separate warm-start objective. Under joint improvement, the first constrained subproblem enters the positive-surplus region. Entropy regularization yields a closed-form opponent, and the KKT multipliers recover inverse-surplus objective weights. A low-dimensional dual update and partition-free pairwise log-ratio regression then implement each policy step from binary comparisons without fitting scalar rewards. Simultaneous confidence intervals certify accepted neural stages, and the convergence analysis covers exact last iterates as well as approximate updates with explicit sampling and regression errors.

Overall, NBPO provides an end-to-end separation between objective-wise games that preserve potentially cyclic feedback and a reference-relative bargaining rule that selects, optimizes, and certifies one multi-objective policy.

2 Background

General preference models. A prompt x is drawn from ρ, and a policy $\pi(\cdot \mid x)$ is a distribution over a finite response set $\mathcal{Y}_x$. For the finite-dimensional population analysis, we take the support of ρ and every $\mathcal{Y}_x$ to be finite. Let $\Pi = \prod_{x\in\mathrm{supp}(\rho)} \Delta(\mathcal{Y}_x)$ denote the contextual policy simplex. Objective $k \in [K]$ is represented by a pairwise preference oracle $P_k(y \succ z \mid x) \in [0,1]$ satisfying

$$P_k(y \succ z \mid x) + P_k(z \succ y \mid x) = 1, \qquad P_k(y \succ y \mid x) = \tfrac{1}{2}. \tag{1}$$

Define the centered payoff

$$A_k(y,z \mid x) = P_k(y \succ z \mid x) - \tfrac{1}{2}, \qquad A_k(y,z \mid x) = -A_k(z,y \mid x), \tag{2}$$

and the objective-k expected centered pairwise payoff of policy π against comparator policy ν,

$$g_k(\pi, \nu) = \mathbb{E}_{x\sim\rho,\, y\sim\pi(\cdot|x),\, z\sim\nu(\cdot|x)} A_k(y, z \mid x). \tag{3}$$

Thus, $g_k(\pi, \nu) > 0$ means that π is preferred to ν on average under objective k, while $g_k(\pi, \nu) = 0$ corresponds to an average tie. No scalar reward representation or transitivity assumption is imposed.

Bargaining problems. A K-dimensional bargaining problem is a feasible utility set $\mathcal{G} \subseteq \mathbb{R}^K$ and disagreement point $d \in \mathcal{G}$. It is nondegenerate if some $u \in \mathcal{G}$ satisfies $u > d$ componentwise. On a compact, convex, comprehensive problem, the Nash solution is

$$v^{\mathrm{N}} \in \arg \max_{v\in\mathcal{G}, v>d} \sum_{k=1}^{K} \log(v_k - d_k). \tag{4}$$

Here, the optimization variable is the feasible value vector v, and each $v_k - d_k$ is the surplus over the disagreement point. The Nash solution is optimized over feasible value vectors satisfying $v > d$ componentwise (Nash, 1950). Its derivative with respect to coordinate v_k is the inverse surplus, $1/(v_k - d_k)$.

3 Objective-Wise Preference Games

This section defines one scalar value for each objective while preserving its full pairwise preference structure. For objective k, we evaluate the candidate policy π against an adaptive opponent ν that searches for responses against which π performs poorly. The reference policy μ plays two distinct roles: it anchors this comparator here, and it serves as the deployed fallback in the bargaining problem of Section 4.

Definition 1 (Entropy-regularized game value)**.** *For objective k and temperature $\beta_k > 0$, define*

$$V_{k,\beta_k}(\pi) = \min_{\nu} \left\{ g_k(\pi, \nu) + \beta_k \mathbb{E}_{x\sim\rho} \mathrm{KL}\left(\nu(\cdot \mid x) \| \mu(\cdot \mid x)\right) \right\}. \tag{5}$$

The first term is the centered preference margin of π against ν. Because ν minimizes this margin, it emphasizes responses that expose weaknesses of the current policy, while the KL term keeps the comparator near μ. Hence β_k controls the locality of the evaluation: a large β_k keeps ν close to μ, whereas $\beta_k \downarrow 0$ recovers the unregularized worst-comparator value $\min_\nu g_k(\pi, \nu)$.

For a fixed opponent response z, define

$$r_{k,\pi}(z \mid x) = \mathbb{E}_{y\sim\pi(\cdot|x)} A_k(y, z \mid x). \tag{6}$$

Thus $r_{k,\pi}(z \mid x)$ is the average centered margin of π against z; smaller values correspond to stronger challenges to the current policy. The KL-regularized inner problem then has the closed-form solution

$$\nu^{\star}_{k,\pi}(z \mid x) = \frac{\mu(z \mid x) \exp\{-r_{k,\pi}(z \mid x)/\beta_k\}}{\mathbb{E}_{z'\sim\mu(\cdot|x)} \exp\{-r_{k,\pi}(z' \mid x)/\beta_k\}}, \tag{7}$$

$$V_{k,\beta_k}(\pi) = -\beta_k \mathbb{E}_{x\sim\rho} \log \mathbb{E}_{z\sim\mu(\cdot|x)} \exp\{-r_{k,\pi}(z \mid x)/\beta_k\}. \tag{8}$$

Equation (7) therefore upweights responses with smaller margins, and Equation (8) is the corresponding soft minimum of the policy's pairwise margins.

Proposition 1 (Structure)**.** *For every k and $\beta_k \geq 0$, V_{k,β_k} is continuous and concave in π. For $\beta_k > 0$, it is differentiable, with objective-wise policy gradient represented by*

$$q_{k,\pi}(y \mid x) = \mathbb{E}_{z\sim\nu^{\star}_{k,\pi}(\cdot|x)} A_k(y, z \mid x). \tag{9}$$

The quantity $q_{k,\pi}(y \mid x)$ is the expected centered margin of response y against the current adaptive opponent. It therefore measures the local effect of increasing the probability of y on the objective-k game value. Concavity will be used to construct the bargaining problem in Section 4, while this gradient characterization will yield the policy update in Section 5. This distinction matters when pairwise preferences contain structure that is invisible from the reference alone. The following example shows that two policies can have the same fixed-reference performance while having fundamentally different vulnerability to other responses.

Why a fixed reference can miss cyclic structure. Consider one prompt and one objective. Let δ_y denote the deterministic policy that always outputs response y, and let the reference be $\mu = \delta_r$. Suppose A, B, and C each beat r with probability 0.75, but form the deterministic cycle $A \succ B$, $B \succ C$, and $C \succ A$. The pure policy δ_A and the uniform mixture $\pi_{\mathrm{mix}} = (\delta_A + \delta_B + \delta_C)/3$ have the same fixed-reference score, $P(\pi \succ r) - 1/2 = 0.25$. Their worst-case game values differ: at $\beta = 0$, response C defeats δ_A, so $V_0(\delta_A) = -0.5$, whereas the uniform mixture ties every response in the cycle, so $V_0(\pi_{\mathrm{mix}}) = 0$. Since $V_0(\mu) = -0.25$, their improvements over the fallback are -0.25 and 0.25, respectively. A fixed reference therefore cannot distinguish a fully exploitable pure response from a balanced mixture once both have the same reference win rate. Appendix A.2 gives the full calculation and formalizes the limiting relation to the fixed-reference precursor.

4 Nash Bargaining over Game-Value Improvements

This section converts the objective-wise game values from Section 3 into a bargaining problem over improvements relative to the reference. We state the feasibility condition under which this problem is well defined, characterize the resulting Nash solution, and compare its population guarantees with utilitarian and maxmin aggregation. The reference defines one fallback value and one improvement (the bargaining *surplus*) per objective:

$$d_k = V_{k,\beta_k}(\mu), \qquad s_k(\pi) = V_{k,\beta_k}(\pi) - d_k. \tag{10}$$

A positive surplus means that π has a higher regularized game value than the deployed reference for objective k.

Assumption 1 (Joint improvement and support)**.** *The reference μ has full support, and some policy $\bar{\pi}$ satisfies $s_k(\bar{\pi}) > 0$ for every k.*

The assumption can fail when the reference is already optimal for one objective or when no policy jointly improves the full objective set. Rather than treating positivity as a separate preprocessing problem, we expose it directly through one auxiliary surplus variable per objective:

$$\begin{aligned}(\pi^{\mathrm{N}}, u^{\mathrm{N}}) \in \arg \max_{\pi,\, u \in \mathbb{R}^K_{++}} \quad & \sum_{k=1}^{K} \log u_k \\ \text{s.t.} \quad & u_k \le s_k(\pi), \qquad k \in [K], \\ & \pi \in \Pi.\end{aligned} \tag{11}$$

Because every s_k is concave, each constraint is the hypograph of a concave function, and (11) is a convex optimization problem in the standard sense of maximizing a concave objective over a convex set. It is not a linear program at finite temperature because V_{k,β_k} contains the log-sum-exp in (8). The objective is strictly increasing in every u_k, so every optimum satisfies $u_k = s_k(\pi)$; hence (11) is equivalent to maximizing

$$F(\pi) = \sum_{k=1}^{K} \log s_k(\pi) \qquad \text{over} \qquad \Pi_+ = \{\pi : s_k(\pi) > 0\ \forall k\}. \tag{12}$$

We do not add an objective-level KL term. KL appears only in the opponent definition and as the proximal geometry used to solve the same constrained objective.

Let $V_{\boldsymbol{\beta}}(\pi) = (V_{1,\beta_1}(\pi), \ldots, V_{K,\beta_K}(\pi))$ and define the comprehensive individually rational hull

$$\mathcal{G}_{\boldsymbol{\beta}} = \left\{ u \in \mathbb{R}^K : d \le u \le V_{\boldsymbol{\beta}}(\pi) \text{ componentwise for some } \pi \right\}. \tag{13}$$

The next result verifies that these policy-induced game values form a valid classical bargaining problem; this is the step that justifies applying the Nash solution and its standard outcome guarantees.

Theorem 1 (Bargaining outcome)**.** *Under Assumption 1, $\mathcal{G}_{\boldsymbol{\beta}}$ is compact, convex, comprehensive, and nondegenerate. Maximizing* (11) *is equivalent to applying the classical Nash solution to* $(\mathcal{G}_{\boldsymbol{\beta}}, d)$*. Every optimizer strictly improves every objective over the fallback and is Pareto optimal in $V_{\boldsymbol{\beta}}$, and all optimizers induce the same game-value vector.*

The policy itself can remain nonunique because distinct response distributions may realize the same game-value vector. Beyond these core outcome guarantees, two further properties are especially relevant when objectives use different numerical scales and when improvements must be balanced across judges.

Theorem 2 (Proportional fairness and scale covariance)**.** *Let $s^{\mathrm{N}} = s(\pi^{\mathrm{N}})$ be the unique surplus vector selected by* (11)*. Every feasible policy $\pi \in \Pi_+$ satisfies*

$$\sum_{k=1}^{K} \frac{s_k(\pi) - s_k^{\mathrm{N}}}{s_k^{\mathrm{N}}} \le 0. \tag{14}$$

Moreover, for arbitrary $c_k > 0$ and $b_k \in \mathbb{R}$, replacing (V_k, d_k) by $(c_k V_k + b_k, c_k d_k + b_k)$ leaves the policy optimizer set unchanged.

To clarify why these guarantees favor Nash bargaining over common aggregation alternatives, Table 1 compares their population-level properties under the same joint-positive-surplus setting. Each entry is a guarantee for every global population optimizer; positive-weight utilitarian uses fixed weights $w_k > 0$, and affine covariance refers to jointly transforming each game-value coordinate and its fallback.

Table 1: Population guarantees under joint positive surplus.

rule	strict IR	Pareto optimal	unique value	scale covariance
positive-weight utilitarian	no	yes	no	no
absolute maxmin	no	no	no	no
surplus maxmin	yes	no	no	no
Nash bargaining	yes	yes	yes	yes

Appendix A.5 gives explicit finite-game counterexamples for the other rules. Appendix A.4 also distinguishes covariance of the utility representation from rescaling raw preference labels at fixed temperature.

5 Nash Bargaining Preference Optimization

This section solves (11) without a separate feasibility objective. The key is an implicit entropy-proximal step whose constraints enforce positive surplus and whose dual variables recover the Nash inverse-surplus weights.

5.1 Constrained entropy-proximal update

For an interior policy, Proposition 1 gives

$$\nabla F(\pi) = \sum_{k=1}^{K} \frac{q_{k,\pi}}{s_k(\pi)}, \qquad q_{k,\pi}(y \mid x) = \mathbb{E}_{z \sim \nu^{\star}_{k,\pi}(\cdot \mid x)} A_k(y, z \mid x). \tag{15}$$

This explicit gradient is undefined at the reference because $s_k(\mu) = 0$. We therefore do not evaluate it there. Let $D(\pi \| \pi_t) = \mathbb{E}_{x \sim \rho(\cdot)} \mathrm{KL}(\pi(\cdot \mid x) \| \pi_t(\cdot \mid x))$. Starting from $\pi_0 = \mu$, stage t solves

$$\begin{aligned} (\pi_{t+1}, u_{t+1}) \in \arg\max_{\pi,\, u \in \mathbb{R}^K_{++}} \quad & \sum_{k=1}^{K} \log u_k - \frac{1}{\eta_t} D(\pi \| \pi_t) \\ \text{s.t.} \quad & u_k \le s_k(\pi), \qquad k \in [K], \\ & \pi \in \Pi, \end{aligned} \tag{16}$$

where $\eta_t > 0$ is an algorithmic step size. The KL term is proximal, not part of the bargaining objective. Under Assumption 1, (16) is feasible for every center π_t, including μ, and its optimum has $u_{t+1,k} > 0$. Thus the first update itself enters the positive-surplus domain.

The Lagrangian of (16) is

$$\mathcal{L}_t(\pi, u, \lambda) = \sum_k \log u_k + \sum_k \lambda_k \big(s_k(\pi) - u_k\big) - \frac{1}{\eta_t} D(\pi \| \pi_t), \qquad \lambda \ge 0. \tag{17}$$

Maximizing over $u_k > 0$ gives $u_k = 1/\lambda_k$ and the K-dimensional dual

$$\phi_t(\lambda) = -\sum_{k=1}^{K} \log \lambda_k - K + \max_{\pi\in\Pi} \left\{ \sum_{k=1}^{K} \lambda_k s_k(\pi) - \frac{1}{\eta_t} D(\pi\|\pi_t) \right\}, \qquad \lambda \in \mathbb{R}^K_{++}, \tag{18}$$

$$\nabla_k \phi_t(\lambda) = s_k\big(\pi_t(\lambda)\big) - \frac{1}{\lambda_k}, \tag{19}$$

where $\pi_t(\lambda)$ is the unique maximizer in (18).

Proposition 2 (Endogenous inverse-surplus weights)**.** *Under Assumption 1, strong duality holds for* (16)*. At its solution,*

$$u_{t+1,k} = s_k(\pi_{t+1}), \qquad \lambda_{t+1,k} = \frac{1}{u_{t+1,k}} = \frac{1}{s_k(\pi_{t+1})}. \tag{20}$$

Hence the inverse-surplus weights are KKT multipliers of the positive-surplus constraints rather than clipped weights evaluated before the update.

5.2 Sampled dual solve and partition-free regression

For fixed λ, the policy maximizer in (18) is characterized by an implicit pairwise log-ratio equation. For a full-support proposal over response pairs, draw y, y' and, for each objective, draw $z_k \sim \nu^\star_{k,\bar\pi}(\cdot \mid x)$ for the current inner candidate $\bar\pi$. Query independent binary outcomes and define

$$Z_{t,k} = B_k(y, z_k) - B_k(y', z_k). \tag{21}$$

Then $\mathbb{E}[Z_{t,k} \mid y, y', z_k] = A_k(y, z_k) - A_k(y', z_k)$. Let

$$h_t(\pi; y, y') = \log \frac{\pi(y \mid x)}{\pi_t(y \mid x)} - \log \frac{\pi(y' \mid x)}{\pi_t(y' \mid x)}. \tag{22}$$

At the fixed point $\bar\pi = \pi_t(\lambda)$, the population loss

$$\mathcal{J}_{t,\lambda}(\pi) = \mathbb{E}\left[h_t(\pi; y, y') - \eta_t \sum_{k=1}^{K} \lambda_k Z_{t,k} \right]^2 \tag{23}$$

has $\pi_t(\lambda)$ as its unique full-support minimizer. In practice we alternate opponent reweighting and regression until the fixed-point residual stabilizes. For numerical conditioning one may use $\tilde{\lambda}_k = \lambda_k / \sum_j \lambda_j$ and multiply the regression coefficient by $\sum_j \lambda_j$; the target in (23) is then unchanged exactly.

The dual is only K-dimensional. Given a training estimate $\widehat{s}(\pi_t(\lambda^{(m)}))$, one projected dual step is

$$\lambda^{(m+1)} = \Pi_\Lambda \Big[\lambda^{(m)} - \gamma_m \left(\widehat{s}\big(\pi_t(\lambda^{(m)})\big) - (\lambda^{(m)})^{-1} \right) \Big], \tag{24}$$

where the inverse is elementwise and $\Lambda = [\lambda_{\min}, \lambda_{\max}]^K$ is a broad numerical box. If the box contains the KKT multipliers, projection does not change the solution; a boundary hit triggers expansion rather than silent clipping. With exact weighted-policy solves, ϕ_t is strongly convex and smooth on Λ, so projected dual descent converges geometrically for a sufficiently small fixed γ_m; Appendix A.6 gives the argument.

Estimating a surplus requires sampling because (8) contains expectations over prompts, learner responses, and reference comparators. Training uses a consistent plug-in estimate from sampled prompts, M reference responses per prompt, and L learner comparisons per reference response; its finite-sample bias is treated as part of the inexact-gradient error. Acceptance uses an independent or preallocated sealed batch and the simultaneous nested intervals in Appendix B. The valid surplus interval is

$$\underline{s}_k(\pi) = \underline{V}_k(\pi) - \overline{V}_k(\mu), \qquad \overline{s}_k(\pi) = \overline{V}_k(\pi) - \underline{V}_k(\mu). \tag{25}$$

For any already accepted current policy, a valid lower bound on the welfare change is

$$\underline{\Delta F}_t = \sum_k \log \underline{s}_k(\widetilde{\pi}_{t+1}) - \sum_k \log \overline{s}_k(\pi_t). \tag{26}$$

A candidate is accepted only if all lower surpluses are positive and, once an accepted current policy exists, $\underline{\Delta F}_t \geq 0$.

Algorithm 1 Constrained Nash Bargaining Preference Optimization

Require: Reference μ, preference oracles $P_{1:K}$, temperatures $\beta_{1:K}$, proximal steps $\eta_{0:T-1}$, dual steps $\gamma_{0:M-1}$, confidence δ.

Estimate $d_k = V_{k,\beta_k}(\mu)$ for training and cache independent simultaneous reference intervals for certification.
Set $\pi_0 \leftarrow \mu$, initialize positive dual weights, and set $\mathsf{accepted} \leftarrow \mathrm{false}$.
for $t = 0, \dots, T-1$ **do**
- Initialize $\lambda^{(0)}$ from the previous stage (or **1** at $t = 0$).
- **for** $m = 0, \dots, M-1$ **do**
  - Approximate $\pi_t(\lambda^{(m)})$ by alternating regularized-opponent reweighting and regression with (23).
  - Estimate training surpluses $\widehat{s}_k(\pi_t(\lambda^{(m)}))$ by nested sampling.
  - Update $\lambda^{(m+1)}$ by (24).
- **end for**
- Set the fitted candidate $\widetilde{\pi}_{t+1} \leftarrow \pi_t(\lambda^{(M)})$ and certify it on the batch assigned to this check.
- **if** some $\underline{s}_k(\widetilde{\pi}_{t+1}) \leq 0$ **then**
  - Refine the inner solve, increase the assigned sample budget, or set $\pi_{t+1} \leftarrow \pi_t$.
- **else if** $\mathsf{accepted} = \mathrm{true}$ and $\underline{\Delta F}_t < 0$ **then**
  - Refine the inner solve or set $\pi_{t+1} \leftarrow \pi_t$.
- **else**
  - Accept $\pi_{t+1} \leftarrow \widetilde{\pi}_{t+1}$ and set $\mathsf{accepted} \leftarrow \mathrm{true}$.
- **end if**

end for
if $\mathsf{accepted} = \mathrm{false}$ **then**
- **return** "no certified joint game-value improvement under the allocated budget."

end if
return the last accepted policy.

5.3 Last-iterate convergence from the reference

Theorem 3 (Exact proximal last-iterate convergence)**.** *Under Assumption 1, let $\pi^\star$ maximize* (11)*, write $s^\star = s(\pi^\star)$, set $\pi_0 = \mu$, and let* (16) *be solved exactly with arbitrary steps $\eta_t > 0$. Then π_t has full support and belongs to Π_+ for every $t \geq 1$, $F(\pi_{t+1}) \geq F(\pi_t)$ for $t \geq 1$, and, for every $T \geq 1$,*

$$F(\pi^\star) - F(\pi_T) \leq \frac{D(\pi^\star \| \mu)}{\sum_{t=0}^{T-1} \eta_t}. \tag{27}$$

The optimal surplus vector $s^\star$ is unique, and

$$\|s(\pi_T) - s^\star\|_2^2 \leq \frac{2D(\pi^\star \| \mu)}{\sum_{t=0}^{T-1} \eta_t}. \tag{28}$$

In particular, a constant $\eta_t \equiv \eta$ gives $O(1/T)$ last-iterate welfare convergence and $O(T^{-1/2})$ surplus-vector convergence, with no upper bound on η and no positive-surplus initialization.

Appendix A.7 proves the result and gives an inexact extension. Indexing only accepted fitted stages, if their variational residuals are δ_t, their uniform game-gradient errors are ξ_t, and certification makes welfare nondecreasing, the numerator of (27) increases by $\sum_t(\delta_t + 2\eta_t\xi_t)$. Rejected proposals leave the current policy unchanged and are not counted as proximal progress.

6 Experiments

Setup. We evaluate four objectives—helpfulness, honesty, instruction following, and truthfulness—on UltraFeedback. The base and reference policy are `Llama-3.1-8B-Instruct`. A prompted `Qwen3-32B` preference oracle supplies objective-specific training comparisons; each pair is queried in both presentation orders and swap-averaged. Final evaluation uses `Llama-3.3-70B`. Appendix M gives the rubrics and label construction.

Table 2: LLM as a judge results on two backbones. All entries are swap-averaged win rates against that backbone's own reference policy on the benchmark's own prompts, judged by a single pinned open-weights judge, `Llama-3.3-70B-Instruct`, which supplied none of the training preferences. "Alpaca LC" is the length-controlled AlpacaEval variant, "len.-matched" the Arena-Hard win rate restricted to response pairs within a $1.3\times$ length band, and "mean tokens" the mean generated Arena-Hard length; the last two are included because open judges reward verbosity, and across these arms the win rate correlates 0.58 with response length. Within each block every arm starts at $\pi_0 = \mu$, shares the pair set, optimizer and budget, and differs only in its aggregation rule. NBPO is reported at the setting its own Nash product selects and MOPO at τ=0.1; the paragraph below explains both choices, how to read the margins, and why the `Qwen3-8B` block required suppressing that model's reasoning trace.

method	Arena-Hard v2	AlpacaEval 2	Alpaca LC	MT-Bench	len.-matched	mean tokens
Backbone and reference policy: `Llama-3.1-8B-Instruct`						
Base $= \mu = \pi_0$	0.500	0.500	0.500	0.500	0.500	2859
NBPO (ours)	**0.558**	**0.554**	**0.554**	0.553	**0.561**	3034
NBPO, 300-step budget	0.546	0.549	0.530	0.538	0.556	2904
PROSPER/MaxEntBW	0.515	0.546	0.541	**0.560**	0.526	3124
Utilitarian	0.522	0.544	0.548	0.516	0.529	3045
Absolute maxmin	0.532	0.536	0.548	0.525	0.529	3130
Surplus maxmin	0.505	0.532	0.524	0.503	0.536	2978
Uniform-INPO	0.527	0.517	0.535	0.503	0.535	2976
MOPO	0.508	0.512	0.516	0.513	0.495	2862
HT-MNPO (honesty)	0.549	0.528	0.522	0.519	0.549	3006
HT-MNPO (truthfulness)	0.536	0.531	0.517	0.525	0.533	2951
HT-MNPO (helpfulness)	0.530	0.550	0.546	0.494	0.555	3127
HT-MNPO (instr. following)	0.515	0.529	0.533	0.541	0.524	2957
Backbone and reference policy: `Qwen3-8B`*, reasoning trace suppressed*						
Base $= \mu = \pi_0$	0.500	0.500	0.500	0.500	0.500	3215
NBPO (ours)	**0.512**	**0.562**	**0.534**	**0.529**	**0.508**	3402
PROSPER/MaxEntBW	0.505	0.535	0.519	0.497	0.504	3316
Utilitarian	0.509	0.528	0.511	0.497	0.507	3307
Absolute maxmin	0.510	0.531	0.514	0.503	0.503	3316
Surplus maxmin	0.500	0.534	0.523	0.516	0.505	3329
Uniform-INPO	0.491	0.498	0.496	0.481	0.490	3194
MOPO	0.498	0.499	0.497	0.468	0.493	3218
HT-MNPO (honesty)	0.493	0.500	0.503	0.513	0.494	3210
HT-MNPO (truthfulness)	0.503	0.517	0.508	0.528	0.503	3257
HT-MNPO (helpfulness)	0.491	0.497	0.498	0.519	0.497	3215
HT-MNPO (instr. following)	0.491	0.491	0.490	0.509	0.495	3205

Results.

Independence from objective utility scales. The one comparison that does not depend on which panel is chosen is the rescaling test of Theorem 2. Each objective's utility is a von Neumann–Morgenstern payoff, fixed only up to its own positive affine transformation, so a rule whose output moves when one objective is rescaled is reading a unit that carries no preference information. Table 4 multiplies each objective's payoff and temperature by its own factor on the frozen pool—a transformation that leaves every underlying preference unchanged—and recomputes each rule's selection without retraining, decoding or relabelling. NBPO moves 0.01% of pairs, which is the numerical floor of the solver rather than a real deviation, while the utilitarian rule moves 5.9%, PROSPER 6.9% and the two maxmin rules 9.9% and 10.9%. The mechanism is visible in the multipliers: under $\lambda^{(A)}$ the dual moves from $(16.6, 14.5, 15.9, 18.0)$ to $(82.7, 24.1, 15.9, 10.6)$, absorbing the rescaling exactly. This is the empirical counterpart of Theorem 2 and, unlike the surplus comparisons above, it holds by construction rather than by dataset. It is a statement about the selection rule on a frozen pool. Appendix I carries the same rescaling through dual solving and training and finds that the exactness is a property of the solution rather than of any particular run: it survives a numerically re-solved dual only as a margin over the utilitarian control, and at the level of trained policies both rules move less than a change of random seed does, so that stage is currently undecided in either direction.

Table 3: Objective-wise held-out surplus over the reference, grouped by dataset, on the population scale of (2). Each panel reports the minimum and the average surplus across that panel's own objectives: instruction following, truthfulness, honesty and helpfulness on ULTRAFEEDBACK; coverage, faithfulness, conciseness and helpfulness on TL;DR; helpfulness and harmlessness on SAFERLHF. Within a panel every arm starts at $\pi_0 = \mu$, trains on the identical response pool with the identical optimizer budget, and differs only in the weight vector over objectives, so the comparison isolates the aggregation rule. Across panels the magnitudes are not comparable: all three panels use `Llama-3.1-8B-Instruct` as backbone, reference and initialisation, but ULTRAFEEDBACK is scored on a 657-prompt jointly complete panel and the other two on 100 held-out prompts against four anchor samples. The `Qwen3-8B` replication of the ULTRAFEEDBACK panel is in Table 19. The single-objective HT-MNPO corners cannot share rows across panels because each panel has its own objectives, so they are summarised by the corner with the largest and the smallest minimum surplus. Best in each column is bold. Bootstrap intervals and paired significance tests for every entry are in Table 17; per-objective breakdowns are in Table 19.

	ULTRAFEEDBACK		TL;DR		SAFERLHF	
method	min s	avg s	min s	avg s	min s	avg s
NBPO (ours)	**0.039**	**0.063**	−0.001	**0.227**	0.202	0.219
PROSPER/MaxEntBW	−0.051	−0.016	−0.137	0.171	0.232	0.268
Utilitarian	−0.077	−0.046	−0.092	0.218	0.276	0.295
Absolute maxmin	−0.062	−0.024	−0.159	0.160	**0.288**	**0.298**
Surplus maxmin	−0.014	0.028	**0.036**	0.128	0.287	0.294
Uniform-INPO	−0.008	0.028	−0.104	−0.082	−0.154	−0.132
MOPO	0.022	0.028	−0.001	0.061	0.191	0.197
HT-MNPO, best corner	0.022	0.037	−0.020	0.039	−0.098	−0.073
HT-MNPO, worst corner	−0.073	−0.037	−0.158	0.058	−0.100	−0.074

Table 4: Independent objective rescaling on the frozen 7,000-prompt ULTRAFEEDBACK pool. Each entry is the percentage of response pairs whose preferred element flips when objective k's payoff and temperature are both multiplied by λ_k, a transformation that leaves every underlying preference unchanged. Lower is better; 0 is the value implied by Theorem 2. No training, decoding, or judging is repeated. For the constrained solver the dual is re-solved from $\lambda = \mathbf{1}$ on the rescaled pool, exactly as a real run would: under $\lambda_k^{(A)} = (0.2, 0.6, 1.0, 1.7)$ its multipliers move from $(16.6, 14.5, 15.9, 18.0)$ to $(82.7, 24.1, 15.9, 10.6)$, absorbing the rescaling so that the weighted score is left unchanged and 0.01% of pairs flip. Unlike the panel rule it needs no floor on $1/s_k$, because λ is a strictly positive KKT multiplier by construction; the panel rule attains exact invariance here only because its floor does not bind at these rescalings.

selection rule	$\lambda^{(A)}$	$\lambda^{(B)}$	$\lambda^{(C)}$	mean
NBPO (ours)	**0.01**	**0.01**	**0.00**	**0.01**
Utilitarian	5.66	6.31	5.85	5.94
Uniform-INPO	5.66	6.31	5.85	5.94
MOPO	5.66	6.31	5.85	5.94
PROSPER/MaxEntBW	7.35	6.63	6.71	6.90
Surplus maxmin	16.02	13.57	0.00	9.86
Absolute maxmin	16.32	0.00	16.32	10.88

7 RELATED WORK

General pairwise and multi-objective preferences. NLHF and its scalable variants optimize directly against general pairwise preference oracles rather than fitting a scalar reward (Munos et al., 2024; Wu et al., 2025; Zhang et al., 2025; Calandriello et al., 2024; Rosset et al., 2024; Zhou et al., 2025). Blackwell-style methods extend this view to vector-valued feedback and target sets (Blackwell, 1956; Bhatia et al., 2020). PROSPER introduces a maximum-entropy Blackwell winner with a scalable single-policy regression update and an inner worst-case choice over objective weights and opponent policies (Zhang et al., 2026). NBPO instead retains one regularized game value per objective and applies reference-relative Nash bargaining across those values. The contribution is this separation of within-objective preference representation from across-objective aggregation, together with the corresponding constrained optimization, certification, and convergence analysis.

Multi-objective alignment. MODPO, Rewarded Soups, and Panacea expose user-specified trade-offs through scalarization, policy interpolation, or preference conditioning (Zhou et al., 2024; Ramé et al., 2023; Zhong et al., 2024b). MaxMin-RLHF and RMOD optimize a weakest-objective rule, while MOPO designates a primary objective and lower-bound constraints for the others (Chakraborty et al., 2024; Son et al., 2026; Agnihotri et al., 2026). These approaches provide useful control when weights, priorities, or thresholds are available. NBPO addresses a different setting: one compromise policy for a fixed objective panel without prespecified priorities. RACO applies clipped conflict-averse gradient combination to objective-specific preference losses and traces a Pareto frontier for user-specified weights (Chen et al., 2026). It is therefore a relevant frontier method, but does not select a single priority-free compromise.

Social choice and bargaining. Social-choice work argues that alignment should state its aggregation rule explicitly (Conitzer et al., 2024). Multi-party RLHF studies utilitarian, Nash, and leximin welfare over party-specific learned rewards, and separately considers reward-free von Neumann solutions (Zhong et al., 2024a). Nash bargaining has also been used to combine task gradients in multi-task learning (Navon et al., 2022). Our setting applies the Nash rule to improvements in objective-wise game values derived directly from separate pairwise preference oracles.

8 Conclusion and Limitations

NBPO separates two decisions in multi-objective alignment: an objective-wise preference game summarizes potentially cyclic pairwise feedback, and Nash bargaining selects one compromise across the resulting game values. Under joint improvement, the selected outcome is strictly above the reference in every objective, Pareto optimal, unique in game-value space, proportionally fair, and covariant to independent positive affine utility transformations. The constrained entropy-proximal formulation starts directly from the reference, and its KKT multipliers yield the inverse-surplus weights used by regularized-opponent sampling and pairwise regression. Theorem 3 gives last-iterate convergence without requiring a separately constructed interior policy, while Appendix A.2 records the fixed-reference precursor as a limiting case rather than making it a main contribution.

The guarantees apply to the selected game-value coordinates, not to every prompt, every pairwise comparison, or an external safety metric. The objective panel, reference policy, and opponent temperature remain modeling choices, and joint positive improvement can be infeasible. Worst-opponent evaluation can favor conservative mixtures and increases sampling cost. Finally, the current completed comparison contains one matched training run per method; the observed ordering must be replicated across independent seeds before it can support a performance claim.

AI use statement

In this work, generative AI tools were used to critique the problem formulation, assist with mathematical derivations and proof presentation, propose experimental checks, and edit manuscript text. The authors are responsible for independently verifying every theorem, citation, implementation, empirical result, and disclosure in the final submission.

Ethics statement

This work studies aggregation rules for language-model alignment, including harmlessness as one objective. Improvement in a game value is not an absolute safety certificate. Safety-critical deployment should retain hard constraints and independent safety evaluations in addition to the bargaining objective.

Reproducibility statement

Algorithm 1 specifies the constrained training procedure. Appendices A–B provide the proximal-dual derivation, convergence proof, finite-sample certificate, and approximate-update analysis. Appendix D gives detailed results and the matched evaluation protocol. A submission package should include anonymized code, exact prompts, random seeds, model checkpoints, opponent pools, dual residuals, and per-stage confidence intervals.

References

Akhil Agnihotri, Rahul Jain, Deepak Ramachandran, and Zheng Wen. Multi-objective preference optimization: Improving human alignment of generative models. In *International Conference on Machine Learning*, 2026.

Kush Bhatia, Ashwin Pananjady, Peter L. Bartlett, Anca D. Dragan, and Martin J. Wainwright. Preference learning along multiple criteria: A game-theoretic perspective. In *Advances in Neural Information Processing Systems*, volume 33, pp. 7413–7424, 2020.

David Blackwell. An analog of the minimax theorem for vector payoffs. *Pacific Journal of Mathematics*, 6(1):1–8, 1956.

Ralph Allan Bradley and Milton E. Terry. Rank analysis of incomplete block designs: I. the method of paired comparisons. *Biometrika*, 39(3/4):324–345, 1952.

Daniele Calandriello, Daniel Guo, Rémi Munos, Mark Rowland, Yunhao Tang, Bernardo Avila Pires, Pierre Harvey Richemond, Charline Le Lan, Michal Valko, Tianqi Liu, et al. Human alignment of large language models through online preference optimisation. *arXiv preprint arXiv:2403.08635*, 2024.

Souradip Chakraborty, Jiahao Qiu, Hui Yuan, Alec Koppel, Furong Huang, Dinesh Manocha, Amrit Singh Bedi, and Mengdi Wang. Maxmin-rlhf: Alignment with diverse human preferences. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, 2024.

Peter Chen, Xiaopeng Li, Xi Chen, and Tianyi Lin. Reward-free alignment for conflicting objectives. In *Proceedings of the 43rd International Conference on Machine Learning*, 2026.

Paul F. Christiano, Jan Leike, Tom Brown, Miljan Martic, Shane Legg, and Dario Amodei. Deep reinforcement learning from human preferences. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

Vincent Conitzer, Rachel Freedman, Jobst Heitzig, Wesley H. Holliday, Bob M. Jacobs, Nathan Lambert, Milan Mossé, Eric Pacuit, Stuart Russell, Hailey Schoelkopf, Emanuel Tewolde, and William S. Zwicker. Position: Social choice should guide AI alignment in dealing with diverse human feedback. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, 2024.

Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963.

Rémi Munos, Michal Valko, Daniele Calandriello, Mohammad Gheshlaghi Azar, Mark Rowland, Daniel Guo, Yunhao Tang, Matthieu Geist, Thomas Mesnard, Andrea Michi, et al. Nash learning from human feedback. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, 2024.

John F Nash. The bargaining problem. *Econometrica*, 18(2):155–162, 1950.

Aviv Navon, Aviv Shamsian, Idan Achituve, Haggai Maron, Kenji Kawaguchi, Gal Chechik, and Ethan Fetaya. Multi-task learning as a bargaining game. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, 2022.

Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, volume 35, pp. 27730–27744, 2022.

Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

Alexandre Ramé, Guillaume Couairon, Corentin Dancette, Jean-Baptiste Gaya, Mustafa Shukor, Laure Soulier, and Matthieu Cord. Rewarded soups: Towards pareto-optimal alignment by interpolating weights fine-tuned on diverse rewards. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

Corby Rosset, Ching-An Cheng, Arindam Mitra, Michael Santacroce, Ahmed Awadallah, and Tengyang Xie. Direct nash optimization: Teaching language models to self-improve with general preferences. *arXiv preprint arXiv:2404.03715*, 2024.

Seongho Son, William Bankes, Sangwoong Yoon, Shyam Sundhar Ramesh, Xiaohang Tang, and Ilija Bogunovic. Robust multi-objective controlled decoding of large language models. In *International Conference on Learning Representations*, 2026.

Yue Wu, Zhiqing Sun, Huizhuo Yuan, Kaixuan Ji, Yiming Yang, and Quanquan Gu. Self-play preference optimization for language model alignment. In *International Conference on Learning Representations*, 2025.

Jiahao Zhang, Lujing Zhang, Keltin Grimes, Zhuohao Yu, Gokul Swamy, and Zhiwei Steven Wu. Back to blackwell: Closing the loop on intransitivity in multi-objective preference fine-tuning. *arXiv preprint arXiv:2602.19041*, 2026.

Yuheng Zhang, Dian Yu, Baolin Peng, Linfeng Song, Ye Tian, Mingyue Huo, Nan Jiang, Haitao Mi, and Dong Yu. Iterative nash policy optimization: Aligning LLMs with general preferences via no-regret learning. In *The Thirteenth International Conference on Learning Representations*, 2025. URL `https://openreview.net/forum?id=Pujt3ADZgI`.

Huiying Zhong, Zhun Deng, Weijie J. Su, Zhiwei Steven Wu, and Linjun Zhang. Provable multi-party reinforcement learning with diverse human feedback. *arXiv preprint arXiv:2403.05006*, 2024a.

Yifan Zhong, Chengdong Ma, Xiaoyuan Zhang, Ziran Yang, Haojun Chen, Qingfu Zhang, Siyuan Qi, and Yaodong Yang. Panacea: Pareto alignment via preference adaptation for LLMs. In *Advances in Neural Information Processing Systems*, 2024b.

Runlong Zhou, Maryam Fazel, and Simon S. Du. Extragradient preference optimization (EGPO): Beyond last-iterate convergence for nash learning from human feedback. In *Conference on Language Modeling*, 2025.

Zhanhui Zhou, Jie Liu, Jing Shao, Xiangyu Yue, Chao Yang, Wanli Ouyang, and Yu Qiao. Beyond one-preference-fits-all alignment: Multi-objective direct preference optimization. In *Findings of the Association for Computational Linguistics: ACL 2024*, 2024.

A Proofs and Additional Theory

Throughout the proofs, response sets are finite. The contextual case uses the ρ-weighted product of policy simplices and the ρ-weighted KL divergence. We therefore write most curvature calculations for one prompt; the contextual extension is block diagonal.

A.1 Proof of Proposition 1

For fixed ν, the objective in (5) is affine in π. A pointwise infimum of affine functions is concave. Compactness of the finite policy simplices and continuity of the payoff imply continuity.

For $\beta_k > 0$, the minimization separates over prompts. Fix x and abbreviate $r(z) = r_{k,\pi}(z \mid x)$. The Lagrangian for

$$\min_{\nu \in \Delta(\mathcal{Y}_x)} \sum_z \nu(z) r(z) + \beta_k \sum_z \nu(z) \log \frac{\nu(z)}{\mu(z)}$$

gives $\log(\nu(z)/\mu(z)) = -r(z)/\beta_k - c_x$. Normalization yields (7), and substitution gives (8). The minimizer is unique because KL is strictly convex on the support of μ. Danskin's theorem therefore gives

$$DV_{k,\beta_k}(\pi)[h] = \mathbb{E}_x \sum_y h(y \mid x) \mathbb{E}_{z \sim \nu^\star_{k,\pi}(\cdot|x)} A_k(y, z \mid x),$$

where $z \sim \nu^\star_{k,\pi}$, which is (9).

A.2 Connection to the Fixed-Reference Precursor

Define the fixed-reference improvement

$$a_k(\pi) = g_k(\pi, \mu) = P_k(\pi \succ \mu) - \tfrac{1}{2}. \tag{29}$$

Skew-symmetry gives $a_k(\mu) = 0$.

Full cyclic example. For one prompt, let $\mu = \delta_r$, where δ_y is the deterministic policy that always outputs y. Suppose A, B, and C each beat r with probability 0.75 and form the deterministic cycle $A \succ B$, $B \succ C$, and $C \succ A$. Both $\pi_{\text{pure}} = \delta_A$ and $\pi_{\text{mix}} = (\delta_A + \delta_B + \delta_C)/3$ have fixed-reference improvement $0.75 - 0.5 = 0.25$. At temperature zero, however, C defeats π_{pure} deterministically, so $V_0(\pi_{\text{pure}}) = -0.5$. Against each of A, B, C, the mixture wins with probability $(1/2 + 0 + 1)/3 = 1/2$, and it beats r with probability 0.75, so $V_0(\pi_{\text{mix}}) = 0$. The reference loses to each cycle response with probability 0.75, giving $V_0(\mu) = 0.25 - 0.5 = -0.25$. Thus

	$a(\pi)$	$V_0(\pi)$	$V_0(\pi) - V_0(\mu)$
π_{pure}	0.25	-0.50	-0.25
π_{mix}	0.25	0	0.25

The fixed reference collapses two policies with different exploitability to the same score; the objective-wise game value retains the distinction.

Theorem 4 (Fixed-reference and worst-opponent limits)**.** *For every policy π and $\beta_k > 0$,*

$$0 \le a_k(\pi) - V_{k,\beta_k}(\pi) \le \frac{1}{8\beta_k}. \tag{30}$$

If $d_{k,\beta_k} = V_{k,\beta_k}(\mu)$ and $s_{k,\beta_k}(\pi) = V_{k,\beta_k}(\pi) - d_{k,\beta_k}$, then

$$\sup_\pi |s_{k,\beta_k}(\pi) - a_k(\pi)| \le \frac{1}{8\beta_k}. \tag{31}$$

Consequently, $s_{k,\beta_k}(\pi) \to a_k(\pi)$ uniformly as $\beta_k \to \infty$. If μ has full support on the finite response set, then

$$V_{k,\beta_k}(\pi) \longrightarrow \min_\nu g_k(\pi, \nu) \qquad \text{as } \beta_k \downarrow 0. \tag{32}$$

Corollary 1 (Consistency with the fixed-reference precursor)**.** *Suppose some policy satisfies $a_k(\pi) > 0$ for every objective. For any sequence of temperature vectors with $\min_k \beta_k \to \infty$, every accumulation point of finite-temperature Nash optimizers maximizes $\sum_k \log a_k(\pi)$. Their fixed-reference improvement vectors converge to the unique utility vector selected by the precursor bargaining problem.*

A large-temperature expansion also gives

$$V_{k,\beta_k}(\pi) = a_k(\pi) - \frac{1}{2\beta_k} \mathbb{E}_x \operatorname{Var}_{z\sim\mu} \left[r_{k,\pi}(z \mid x) \right] + O(\beta_k^{-2}), \tag{33}$$

so finite temperature adds a correction that depends on how unevenly the current policy performs across reference responses.

Proof of Theorem 4. Fix a prompt and let $R = r_{k,\pi}(Z \mid x)$ for $Z \sim \mu(\cdot \mid x)$. Since $A_k \in [-1/2, 1/2]$, also $R \in [-1/2, 1/2]$. Write

$$V_x = -\beta_k \log \mathbb{E} e^{-R/\beta_k}.$$

Jensen's inequality gives $\log \mathbb{E} e^{-R/\beta_k} \ge -\mathbb{E}R/\beta_k$, hence $V_x \le \mathbb{E}R$. Hoeffding's lemma gives

$$\log \mathbb{E} \exp\left\{ -\frac{R - \mathbb{E}R}{\beta_k} \right\} \le \frac{1}{8\beta_k^2}.$$

Therefore $V_x \ge \mathbb{E}R - 1/(8\beta_k)$. Averaging over x proves (30) because $\mathbb{E}_x \mathbb{E}R = g_k(\pi, \mu) = a_k(\pi)$.

Apply the same argument at $\pi = \mu$. Since $a_k(\mu) = g_k(\mu, \mu) = 0$, there are errors $e_\pi, e_\mu \in [0, 1/(8\beta_k)]$ such that

$$V_{k,\beta_k}(\pi) = a_k(\pi) - e_\pi, \qquad V_{k,\beta_k}(\mu) = -e_\mu.$$

Thus $s_{k,\beta_k}(\pi) - a_k(\pi) = e_\mu - e_\pi$, proving (31).

For the zero-temperature limit, at each prompt

$$-\beta_k \log \sum_z \mu(z \mid x) e^{-r(z)/\beta_k} \longrightarrow \min_{z:\mu(z|x)>0} r(z).$$

Full support makes this the minimum over all responses, which equals $\min_\nu g_k(\pi, \nu)$ after averaging over prompts.

Proof of Corollary 1. Let $\epsilon_n = \max_k (8\beta_{n,k})^{-1} \to 0$ and let $s_{n,k}$ denote the corresponding game surplus. Theorem 4 gives $\sup_\pi |s_{n,k}(\pi) - a_k(\pi)| \le \epsilon_n$. Choose a policy $\bar\pi$ with $m = \min_k a_k(\bar\pi) > 0$. For all sufficiently large n, $s_{n,k}(\bar\pi) \ge m/2$, so an optimizer π_n satisfies

$$F_n(\pi_n) \ge K \log(m/2).$$

Every positive game surplus is at most one. Hence each coordinate obeys $\log s_{n,k}(\pi_n) \ge F_n(\pi_n)$, and therefore $s_{n,k}(\pi_n) \ge (m/2)^K$. Thus the optimizer sequence remains uniformly away from the surplus boundary. On that compact interior region, $\sum_k \log s_{n,k}$ converges uniformly to $\sum_k \log a_k$. Compactness and the standard argmax argument imply that every accumulation point of π_n maximizes the fixed-reference objective. The fixed-reference bargaining problem has a unique utility vector, so the full sequence of anchor-surplus vectors converges to it.

Finally, the cumulant expansion

$$\log \mathbb{E} e^{tR} = t\mathbb{E}R + \frac{t^2}{2} \operatorname{Var}(R) + O(t^3)$$

with $t = -1/\beta_k$ gives (33); boundedness makes the remainder uniform on the finite simplex.

A.3 Proof of Theorem 1

Continuity on the compact policy simplex bounds $V_{\boldsymbol\beta}(\pi)$, hence $\mathcal{G}_{\boldsymbol\beta}$ is bounded. It is closed: if $u_n \to u$ and $u_n \le V_{\boldsymbol\beta}(\pi_n)$, compactness gives a subsequence $\pi_n \to \pi$, and continuity yields $u \le V_{\boldsymbol\beta}(\pi)$. Comprehensiveness follows immediately from the definition.

For convexity, take $u^i \le V_{\boldsymbol\beta}(\pi_i)$ for $i \in \{1, 2\}$ and $\lambda \in [0, 1]$. Objective-wise concavity gives

$$\lambda u^1 + (1-\lambda)u^2 \le \lambda V_{\boldsymbol\beta}(\pi_1) + (1-\lambda)V_{\boldsymbol\beta}(\pi_2) \le V_{\boldsymbol\beta}(\lambda\pi_1 + (1-\lambda)\pi_2).$$

Thus the convex combination lies in $\mathcal{G}_{\boldsymbol\beta}$. Assumption 1 gives a point strictly above d, so the bargaining problem is nondegenerate.

The Nash objective is coordinatewise increasing. Hence an optimum over $\mathcal{G}_{\boldsymbol\beta}$ can be moved to a dominating attainable vector $V_{\boldsymbol\beta}(\pi)$ without decreasing its value, proving equivalence with (11).

The log objective tends to $-\infty$ when any surplus approaches zero. Since a strictly positive feasible surplus exists, the supremum is attained at a point strictly above disagreement. This proves existence and strict individual rationality.

If a policy π' weakly improved every game value over an optimizer $\pi^\star$ and strictly improved one, then every surplus would weakly increase and one would strictly increase. Strict monotonicity of the logarithm would imply $F(\pi') > F(\pi^\star)$, a contradiction. Hence every optimizer is Pareto optimal.

For uniqueness of the utility outcome, suppose two maximizers π_1, π_2 induce distinct vectors. Concavity gives

$$V_{\boldsymbol\beta}\big(\tfrac{1}{2}\pi_1 + \tfrac{1}{2}\pi_2\big) - d \ge \tfrac{1}{2}s(\pi_1) + \tfrac{1}{2}s(\pi_2)$$

coordinatewise. Since $G(s) = \sum_k \log s_k$ is increasing and strictly concave on $\mathbb{R}^K_{++}$,

$$\begin{aligned} F\big(\tfrac{1}{2}\pi_1 + \tfrac{1}{2}\pi_2\big) &\ge G\big(\tfrac{1}{2}s(\pi_1) + \tfrac{1}{2}s(\pi_2)\big) \\ &> \tfrac{1}{2}G(s(\pi_1)) + \tfrac{1}{2}G(s(\pi_2)), \end{aligned}$$

contradicting optimality. All maximizing policies therefore induce the same game-value vector.

A.4 Proof of Theorem 2 and Payoff Rescaling

Changing the utility coordinates together with their fallback values is distinct from changing the raw preference payoff while keeping the opponent temperature fixed. For $\lambda_k > 0$,

$$V_{\beta_k}^{\lambda_k A_k}(\pi) = \lambda_k V_{\beta_k/\lambda_k}^{A_k}(\pi). \tag{34}$$

Thus exact covariance at the oracle level requires jointly scaling (A_k, β_k) to $(\lambda_k A_k, \lambda_k \beta_k)$, or taking the unregularized limit.

Let $\mathcal{S}_+ = \{s \in \mathbb{R}_{++}^K : d + s \in \mathcal{G}_{\boldsymbol{\beta}}\}$. This set is convex, and s^{N} maximizes $G(s) = \sum_k \log s_k$ over $\mathcal{S}_+$. First-order optimality therefore gives

$$\left\langle \nabla G(s^{\mathrm{N}}), s - s^{\mathrm{N}} \right\rangle = \sum_k \frac{s_k - s_k^{\mathrm{N}}}{s_k^{\mathrm{N}}} \leq 0$$

for every $s \in \mathcal{S}_+$. Taking $s = s(\pi)$ proves (14).

Under $V_k' = c_k V_k + b_k$ and $d_k' = c_k d_k + b_k$, the transformed surplus is $s_k' = c_k s_k$. Therefore

$$\sum_k \log s_k'(\pi) = \sum_k \log s_k(\pi) + \sum_k \log c_k,$$

a policy-independent shift. The optimizer set is unchanged.

For raw payoff scaling,

$$\begin{aligned} V_{\beta_k}^{\lambda_k A_k}(\pi) &= \min_\nu \{\lambda_k g_k^A(\pi, \nu) + \beta_k \mathrm{KL}(\nu \| \mu)\} \\ &= \lambda_k \min_\nu \{g_k^A(\pi, \nu) + (\beta_k/\lambda_k) \mathrm{KL}(\nu \| \mu)\}, \end{aligned}$$

which is (34). Setting the transformed temperature to $\lambda_k \beta_k$ gives $V_{\lambda_k \beta_k}^{\lambda_k A_k} = \lambda_k V_{\beta_k}^{A_k}$ and therefore exact Nash covariance.

A.5 Counterexamples for Alternative Aggregation Rules

Proposition 3 (Guarantees of competing rules)**.** *There are jointly improvable finite pairwise games in which positive-weight utilitarian aggregation selects a policy below the fallback for one objective, and instances in which absolute maxmin violates fallback-relative individual rationality. Surplus maxmin is strictly individually rational under Assumption 1, but its optimizer set can contain a policy Pareto dominated by another optimizer. Every Nash optimizer has the guarantees in Theorem 1.*

A positive-weight utilitarian objective is strictly increasing in every coordinate, so every utilitarian maximizer is Pareto optimal. It need not be individually rational. Absolute maxmin can ignore objective-specific disagreement levels. Surplus maxmin is strictly individually rational under Assumption 1, because a feasible policy has positive minimum surplus, but its flat optimum can contain dominated ties. The following skew-symmetric games make all three failures explicit.

All examples use three responses and the uniform reference $\mu = (1/3, 1/3, 1/3)$. Utilities are unregularized game values $V_k(\pi) = \min_j (\pi^\top A_k)_j$.

Utilitarian failure of individual rationality. Let

$$A_1 = \begin{pmatrix} 0 & -1/2 & -1/2 \\ 1/2 & 0 & -1/2 \\ 1/2 & 1/2 & 0 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} 0 & -1/2 & 1/4 \\ 1/2 & 0 & -1/2 \\ -1/4 & 1/2 & 0 \end{pmatrix}.$$

Then $d = (-1/3, -1/12)$. The policy $(1/4, 1/4, 1/2)$ has surplus $(1/12, 1/48) > 0$, but equal-weight utilitarian optimization has an optimizer $(0, 1/5, 4/5)$ with surplus $(7/30, -1/60)$.

Absolute-maxmin failure of individual rationality. Let

$$A_1 = \begin{pmatrix} 0 & -1/2 & 1/4 \\ 1/2 & 0 & 1/2 \\ -1/4 & -1/2 & 0 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} 0 & 1/4 & -1/2 \\ -1/4 & 0 & 1/2 \\ 1/2 & -1/2 & 0 \end{pmatrix}.$$

Again $d = (-1/3, -1/12)$. The policy $(1/3, 7/15, 1/5)$ has surplus $(1/15, 1/15) > 0$. Absolute maxmin instead has an optimizer $(0, 4/5, 1/5)$ with utility $(-1/10, -1/10)$ and surplus $(7/30, -1/60)$.

A dominated surplus-maxmin tie. Let

$$A_1 = \begin{pmatrix} 0 & -1/2 & 1/2 \\ 1/2 & 0 & 1/2 \\ -1/2 & -1/2 & 0 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} 0 & 0 & 1/2 \\ 0 & 0 & 1/2 \\ -1/2 & -1/2 & 0 \end{pmatrix}.$$

Here $d = (-1/3, -1/6)$. Objective 2 has maximal game value zero for every mixture supported on the first two responses, so no policy can have second surplus above $1/6$. Both

$$\pi_c = (1/3, 2/3, 0), \quad s(\pi_c) = (1/6, 1/6),$$

and

$$\pi_e = (0, 1, 0), \quad s(\pi_e) = (1/3, 1/6),$$

therefore maximize the minimum surplus at $1/6$, but π_e Pareto dominates π_c in game values.

A.6 Dual Derivation and Regression Fixed Point

Assumption 1 supplies a strictly feasible pair for (16): choose $\bar{\pi}$ with $s_k(\bar{\pi}) > 0$ and any $0 < u_k < s_k(\bar{\pi})$. Slater's condition therefore gives strong duality. The Lagrangian is (17). For $\lambda_k > 0$,

$$\max_{u_k > 0} \{\log u_k - \lambda_k u_k\} = -\log \lambda_k - 1, \qquad u_k^\star = \frac{1}{\lambda_k},$$

where the displayed optimizer is the slack variable (not an opponent policy). If any $\lambda_k = 0$, the supremum over u_k is infinite, so the dual domain is $\mathbb{R}^K_{++}$. This proves (18).

For fixed $\lambda > 0$, the term $-D(\pi \| \pi_t)/\eta_t$ is strictly concave. Hence the policy maximizer $\pi_t(\lambda)$ is unique. Because π_t has full support and the derivative of negative KL diverges inward at a zero-probability coordinate, the maximizer also has full support. Danskin's theorem gives

$$\nabla_k \phi_t(\lambda) = s_k(\pi_t(\lambda)) - \frac{1}{\lambda_k},$$

which is (19). KKT stationarity in u_k gives $1/u_k - \lambda_k = 0$. Since $\lambda_k > 0$, complementary slackness makes $u_k = s_k(\pi)$, proving Proposition 2.

Inner dual convergence. Fix a box $\Lambda = [\lambda_{\min}, \lambda_{\max}]^K$ containing a stage dual minimizer. The Hessian of $-\sum_k \log \lambda_k$ is bounded below by $m_\Lambda I$ with $m_\Lambda = \lambda_{\max}^{-2}$, while the maximized term in (18) is convex. Hence ϕ_t is m_Λ-strongly convex on Λ. For $\beta_k > 0$, the game values are smooth; the KL term makes the weighted policy maximizer unique and interior, so the implicit-function theorem and compactness of Λ give an $L_\Lambda < \infty$ such that $\nabla \phi_t$ is L_Λ-Lipschitz. Projection is nonexpansive, and therefore exact projected dual descent with a constant $0 < \gamma < 2m_\Lambda / L_\Lambda^2$ obeys

$$\|\lambda^{(m+1)} - \lambda_t^\star\|_2^2 \le \left(1 - 2\gamma m_\Lambda + \gamma^2 L_\Lambda^2\right) \|\lambda^{(m)} - \lambda_t^\star\|_2^2. \tag{35}$$

Thus the exact inner dual loop converges geometrically. Approximate policy solves and sampled surplus estimates are absorbed into the residual terms in (46).

It remains to justify the regression implementation. Fix λ and abbreviate $\bar{\pi} = \pi_t(\lambda)$. First-order optimality of the weighted policy problem in (18) gives, for each prompt and every response pair in the support,

$$\log \frac{\bar{\pi}(y \mid x)}{\pi_t(y \mid x)} - \log \frac{\bar{\pi}(y' \mid x)}{\pi_t(y' \mid x)} = \eta_t \sum_k \lambda_k \left(q_{k,\bar{\pi}}(y \mid x) - q_{k,\bar{\pi}}(y' \mid x)\right). \tag{36}$$

The conditional mean of $Z_{t,k}$ in (21), with $z_k \sim \nu^\star_{k,\bar{\pi}}$, is exactly the difference on the right-hand side. The squared-loss conditional-mean decomposition therefore makes (36) necessary for every population minimizer of (23). Equality of all pairwise log differences implies that the candidate log density and $\bar{\pi}$ differ only by a prompt-dependent additive constant; normalization proves uniqueness.

A.7 Proof of Theorem 3

Let $h(\pi) = \mathbb{E}_x \sum_y \pi(y \mid x) \log \pi(y \mid x)$, so its Bregman divergence is $D(\pi \| \pi_t)$. Because the objective is increasing in u, the policy part of (16) is equivalently

$$\pi_{t+1} \in \arg\max_{\pi \in \Pi_+} \left\{ F(\pi) - \frac{1}{\eta_t} D(\pi \| \pi_t) \right\}. \tag{37}$$

The feasible set is nonempty by Assumption 1, and $F(\pi) \to -\infty$ as any surplus approaches zero. Thus the maximizer exists in Π_+ even when the center π_t is the reference $\mu \notin \Pi_+$. Because π_t has full support and the remaining objective has finite directional derivatives at an interior surplus point, the inward derivative of $-D(\pi \| \pi_t)$ excludes zero-probability coordinates. Induction therefore gives a full-support, positive-surplus π_{t+1}.

Let g_{t+1} be a supergradient of F at π_{t+1}. First-order optimality of (37) and concavity of F imply, for every $\pi \in \Pi_+$,

$$\begin{aligned} F(\pi) - F(\pi_{t+1}) &\le \langle g_{t+1}, \pi - \pi_{t+1} \rangle_\rho & (38)\\ &\le \frac{1}{\eta_t} \langle \nabla h(\pi_{t+1}) - \nabla h(\pi_t), \pi - \pi_{t+1} \rangle_\rho & (39)\\ &= \frac{1}{\eta_t} \left[D(\pi \| \pi_t) - D(\pi \| \pi_{t+1}) - D(\pi_{t+1} \| \pi_t) \right]. & (40) \end{aligned}$$

For $t \ge 1$, π_t is feasible in the next subproblem. Comparing the optimum with π_t gives

$$F(\pi_{t+1}) - \frac{1}{\eta_t} D(\pi_{t+1} \| \pi_t) \ge F(\pi_t), \tag{41}$$

so the objective values are nondecreasing from π_1 onward.

Set $\pi = \pi^\star$ in (40), multiply by η_t, and sum from $t = 0$ to $T - 1$:

$$\sum_{t=0}^{T-1} \eta_t \big(F(\pi^\star) - F(\pi_{t+1}) \big) \le D(\pi^\star \| \mu). \tag{42}$$

Because the gaps are nonincreasing, the left-hand side is at least $\big(\sum_t \eta_t \big) (F(\pi^\star) - F(\pi_T))$, which proves (27).

For every objective, $V_k(\pi) \in [-1/2, 1/2]$ and $d_k = V_k(\mu) \le 0$ because $\nu = \mu$ is feasible and $g_k(\mu, \mu) = 0$. Hence every positive surplus is at most one. On $(0, 1]^K$, $G(s) = \sum_k \log s_k$ is 1-strongly concave in Euclidean norm. The comprehensive positive-surplus set induced by (13) is convex, and first-order optimality of its unique maximizer $s^\star$ gives

$$G(s^\star) - G(s) \ge \frac{1}{2} \| s - s^\star \|_2^2. \tag{43}$$

Combining this with (27) proves (28).

Approximate sampled stages. Reindex the accepted proposals consecutively, so t below counts accepted proximal stages and a rejected proposal contributes neither a new iterate nor a step size. Let $\widehat{g}_{t+1}$ be the game-gradient estimate induced by the fitted opponents and dual weights. Suppose the fitted policy satisfies, for every $\pi \in \Pi_+$,

$$\eta_t \langle \widehat{g}_{t+1}, \pi - \pi_{t+1} \rangle_\rho \le D(\pi \| \pi_t) - D(\pi \| \pi_{t+1}) - D(\pi_{t+1} \| \pi_t) + \delta_t, \tag{44}$$

and $\| \widehat{g}_{t+1} - g_{t+1} \|_\infty \le \xi_t$. Since the contextual ℓ_1 distance between two policies is at most two, concavity and (44) imply

$$\eta_t \big(F(\pi^\star) - F(\pi_{t+1}) \big) \le D(\pi^\star \| \pi_t) - D(\pi^\star \| \pi_{t+1}) + \delta_t + 2\eta_t \xi_t. \tag{45}$$

If the accepted objective values are nondecreasing, as guaranteed on the simultaneous certification event when (26) is nonnegative, then

$$F(\pi^\star) - F(\pi_T) \le \frac{D(\pi^\star \| \mu) + \sum_{t=0}^{T-1} (\delta_t + 2\eta_t \xi_t)}{\sum_{t=0}^{T-1} \eta_t}. \tag{46}$$

Thus the fitted last iterate converges whenever the cumulative inner-solve and gradient-estimation errors are $o(\sum_t \eta_t)$. The exact theorem is the special case $\delta_t = \xi_t = 0$.

B FINITE-SAMPLE CERTIFICATION OF GAME VALUES

This section supplies the certificate invoked by Algorithm 1. It is deliberately conservative but fully explicit. Tighter empirical-Bernstein or paired bounds can replace it without changing the population objective.

B.1 NESTED ESTIMATOR

Fix at most R policy checks, including the cached reference check. Policies may be chosen adaptively, but check c uses a fresh, independent certification batch, or a sealed batch preallocated to that check before training and not revealed earlier. This independence is necessary; repeatedly reusing the same holdout after observing its outcomes is not covered by the theorem below.

For check c, draw n prompts $x_i \sim \rho$. For each prompt draw M comparators $z_{ij} \sim \mu(\cdot \mid x_i)$. For every objective, comparator, and prompt, draw L independent learner responses $y_{ij\ell} \sim \pi_c(\cdot \mid x_i)$ and binary outcomes

$$B_{k,c,i,j,\ell} \sim \text{Bernoulli}(P_k(y_{ij\ell} \succ z_{ij} \mid x_i)).$$

Define

$$\widehat{r}_{k,c,i,j} = \frac{1}{L}\sum_{\ell=1}^{L}\left(B_{k,c,i,j,\ell} - \tfrac{1}{2}\right), \tag{47}$$

$$\widehat{q}_{k,c,i} = \frac{1}{M}\sum_{j=1}^{M}\exp\{-\widehat{r}_{k,c,i,j}/\beta_k\}. \tag{48}$$

Let

$$\epsilon_{\text{lab}} = \sqrt{\frac{\log(6KRnM/\delta)}{2L}}, \tag{49}$$

$$a_k = e^{-1/(2\beta_k)}, \qquad b_k = e^{1/(2\beta_k)}, \tag{50}$$

$$\epsilon_{\text{cmp},k} = (b_k - a_k)\sqrt{\frac{\log(6KRn/\delta)}{2M}}, \tag{51}$$

$$\epsilon_{\text{pr}} = \sqrt{\frac{\log(6KR/\delta)}{2n}}. \tag{52}$$

For each sampled prompt define

$$\ell_{k,c,i} = -\beta_k \log\min\left\{b_k, e^{\epsilon_{\text{lab}}/\beta_k}\widehat{q}_{k,c,i} + \epsilon_{\text{cmp},k}\right\}, \tag{53}$$

$$u_{k,c,i} = -\beta_k \log\max\left\{a_k, e^{-\epsilon_{\text{lab}}/\beta_k}\widehat{q}_{k,c,i} - \epsilon_{\text{cmp},k}\right\}. \tag{54}$$

Finally set

$$\underline{V}_{k,c} = \max\left\{-\tfrac{1}{2}, \frac{1}{n}\sum_{i=1}^{n}\ell_{k,c,i} - \epsilon_{\text{pr}}\right\}, \tag{55}$$

$$\overline{V}_{k,c} = \min\left\{\tfrac{1}{2}, \frac{1}{n}\sum_{i=1}^{n}u_{k,c,i} + \epsilon_{\text{pr}}\right\}. \tag{56}$$

Theorem 5 (Simultaneous nested game-value intervals)**.** *With probability at least $1-\delta$, simultaneously for every objective $k \in [K]$ and every check $c \in [R]$,*

$$\underline{V}_{k,c} \le V_{k,\beta_k}(\pi_c) \le \overline{V}_{k,c}. \tag{57}$$

Consequently, if one check is the reference policy, the surplus interval in (25) *is simultaneous and valid. If every lower surplus is positive, then the checked policy is feasible for* (11) *with probability at least $1-\delta$. For two checked policies π_a and π_b, if every $\underline{s}_k(\pi_a) > 0$ and $\sum_k \log \underline{s}_k(\pi_a) \ge \sum_k \log \overline{s}_k(\pi_b)$, then $F(\pi_a) \ge F(\pi_b)$ on the same event.*

Table 5: Exact optimization on the frozen ULTRAFEEDBACK response pool ($n = 1{,}372$ complete prompts). Surpluses use the population scale of Eq. (10); higher is better except for the Frank–Wolfe (FW) gap. This finite-pool certificate tests the bargaining rule, not neural generalization.

selection rule	min s_k	mean s_k	$\sum_k \log s_k$	FW gap
NBPO	0.221324	0.235957	-5.783311	9.02e-09
Utilitarian	0.217866	0.236070	-5.785865	9.86e-09

B.2 Proof of Theorem 5

For a fixed sampled comparator, $\widehat{r}$ averages L independent variables in $[-1/2, 1/2]$. Hoeffding's inequality (Hoeffding, 1963) and a union bound over $KRnM$ estimates give

$$|\widehat{r}_{k,c,i,j} - r_{k,\pi_c}(z_{ij} \mid x_i)| \le \epsilon_{\text{lab}} \tag{58}$$

for all indices with failure probability at most $\delta/3$. The statement remains valid for adaptively selected π_c because the data used at check c are independent of the history conditional on which π_c is submitted for certification.

Let

$$q_{k,c,i} = \mathbb{E}_{z\sim\mu(\cdot|x_i)} \exp\{-r_{k,\pi_c}(z \mid x_i)/\beta_k\}$$

and let $q^M_{k,c,i}$ be its empirical mean over the M sampled comparators using the exact r values. Each exponential lies in $[a_k, b_k]$. A second Hoeffding bound and a union bound over KRn prompt-level means give

$$|q^M_{k,c,i} - q_{k,c,i}| \le \epsilon_{\text{cmp},k} \tag{59}$$

with failure probability at most $\delta/3$.

On the intersection of (58) and (59),

$$e^{-\epsilon_{\text{lab}}/\beta_k}\widehat{q}_{k,c,i} - \epsilon_{\text{cmp},k} \le q_{k,c,i} \le e^{\epsilon_{\text{lab}}/\beta_k}\widehat{q}_{k,c,i} + \epsilon_{\text{cmp},k}.$$

Since $q_{k,c,i} \in [a_k, b_k]$ and $-\beta_k \log q$ is decreasing, (53) and (54) bound the exact prompt-level game value.

Each exact prompt-level value lies in $[-1/2, 1/2]$. A third Hoeffding bound over the n prompts, union-bounded over KR checks, gives prompt-generalization error at most ϵ_{pr} with failure probability at most $\delta/3$. Combining the three events proves (57). The surplus statement follows by subtracting the worst compatible endpoints of the candidate and reference intervals.

C Finite-Pool Population Certificate

We freeze 1,500 ULTRAFEEDBACK prompts, four policy responses and four reference responses per prompt, four objective-specific pairwise tensors, $\beta_k = 0.25$, and the reference game values before optimization. The complete intersection contains 1,372 prompts. We optimize (11) directly over one four-simplex per prompt in float64. The reported checkpoint is the converged final iterate, not a reward-selected iterate.

The Nash surplus vector is $(0.229230, 0.258926, 0.221324, 0.234350)$, so every coordinate is strictly above disagreement. Its normalized Frank–Wolfe gap is 9.02×10^{-9}; the corresponding positive supporting weights certify that no policy can improve every game value by more than this residual. A deterministic feasible restart agrees in game-value space within 3.03×10^{-9}. Scaling the four payoff, temperature, and disagreement coordinates by $(0.2, 0.6, 1.0, 1.7)$ gives a scale-corrected surplus residual of 2.78×10^{-17}. The utilitarian control attains a slightly larger mean surplus, while the Nash solution attains the larger log product and minimum surplus. These calculations verify the finite-pool selection rule, but do not close the separate neural-regression error in Theorem 3.

Solution-concept stress test. We additionally preregister 200 new three-objective, five-action cyclic games, conditioned only on the joint-improvement assumption required by Theorem 1. We solve

Table 6: Exact population solutions on 200 preregistered cyclic games. "Defined" is the fraction for which the method returned a feasible policy, IR is the fraction strictly above the common fallback in every objective, and the Pareto gap is measured in the common game-value coordinates. MOPO uses objective 1 as primary and preregistered half-attainable thresholds. Methods optimize different social objectives, so the minimum surplus is diagnostic rather than a universal ranking.

selection rule	defined	strict IR	mean min. surplus	Pareto gap
NBPO	1.000	1.000	0.0289	1.64×10^{-13}
PROSPER/MaxEntBW	1.000	0.665	0.0071	1.86×10^{-13}
Utilitarian	1.000	0.595	-0.0009	1.42×10^{-13}
Surplus maxmin	1.000	1.000	**0.0361**	2.16×10^{-13}
MOPO	0.510	0.035	-0.0887	2.67×10^{-2}
HT single-objective corners	1.000	0.030–0.060	-0.1351–-0.1207	5.95–6.75×10^{-13}

every population problem to numerical convergence and compare the resulting selection rules on the same game tensors. Table 6 separates the properties of the selection rule from neural approximation.

NBPO has zero Nash-welfare regret by construction, satisfies strict IR in every feasible game, and has maximum Pareto gap 4.36×10^{-13}. Under independent coordinate scaling by $(0.25, 0.60, 1.40)$, its maximum policy ℓ_1 drift is 7.01×10^{-7}, whereas PROSPER's mean drift is 0.277. Surplus maxmin obtains the larger minimum surplus, as expected from its objective, while utilitarian obtains the larger mean surplus. The experiment therefore verifies the properties that distinguish the Nash rule; it is not evidence that one neural optimizer dominates every alternative metric.

D Experimental Protocol and Detailed Results

Run matrix. For each dataset and training run, initialize the constrained game-value aggregation rules from the same reference policy and use the same proximal-stage budget. Hold the prompt split, decoding parameters, candidate count, comparator pool, labeled pair set, total preference-query budget, total optimizer budget, and $\beta_{1:K}$ fixed within a seed. Count inner dual iterations, opponent scoring, rejected stages, certification queries, and validation used for hyperparameter selection. PROSPER, fixed-reference precursor, MOPO, and Uniform-INPO are external baselines and should retain their canonical update rules while receiving the same total resource budget.

Independent evaluation. Training labels, utility certification, and final evaluation must use disjoint data. The final game values use an independently generated comparator pool and evaluator families that did not provide training labels. Report failure to certify the first constrained iterate as a result rather than replacing it with a clipped or offset Nash objective.

Primary metrics. For a final release, report V_{k,β_k} and s_k on the theoretical scale of (2), together with the minimum surplus, average game value, Nash welfare, IR rate, fixed-reference win rate, KL to the reference, response length, opponent entropy, and accepted-stage rate. The currently archived summary provides only the implementation-scale quantities $\widetilde{s}_k$ and $\widetilde{F}$ shown below. Rows without a canonical implementation audit are omitted. Nash welfare is undefined when a held-out surplus is nonpositive; do not apply an unreported offset.

D.1 Frozen reporting tables for the matched neural study

The following tables fix the reporting surface for the new matched-budget study before its final evaluation. A dash denotes a cell that has not yet been measured; it must not be imputed from a different model, judge, prompt split, or training budget. The primary panel uses four global criteria on held-out UltraFeedback prompts: instruction following, truthfulness, honesty/calibration, and helpfulness. All methods share one response pool and comparison tensor. PROSPER retains its alternating objective-weight and opponent updates, while MOPO uses instruction following as the preregistered primary objective with fixed lower bounds on the other three objectives. The four HT-MNPO rows are single-objective corner policies rather than candidate compromise policies.

Table 7: Main comparison from the available completed matched run, on the population scale of (2). "min s" is the smallest objective improvement, $F = \sum_k \log s_k$, and "ref. win" is the average conventional win rate against the reference. Higher is better within a panel. Detailed objective values are in Appendix D. PROSPER and MOPO are omitted pending evaluation with their canonical algorithms.

	HelpSteer2			SafeRLHF		
method	min s	F	ref. win	min s	F	ref. win
NBPO (ours)	0.104	−8.011	0.537	0.500	−1.319	0.817
Utilitarian	0.109	−7.931	0.560	0.510	−1.314	0.821
Absolute maxmin	0.129	−6.978	0.577	0.496	−1.325	0.811
Surplus maxmin	0.133	−7.286	0.575	0.497	−1.332	0.817
Fixed-reference precursor	0.134	−7.251	0.565	0.520	−1.278	0.837
Uniform-INPO	0.079	−9.156	0.514	0.495	−1.328	0.804

Table 8: Cross-domain check on TL;DR summarisation. Held-out surpluses on 100 prompts disjoint from training, population scale of (2), judged by the same pinned `Qwen3-32B` oracle and anchored on the same `Llama-3.1-8B-Instruct` reference used throughout. Every arm starts at $\pi_0 = \mu$, trains on the identical eight-response pool with the identical optimizer budget, and differs only in the weight vector over objectives, so the comparison isolates the aggregation rule. The panel is chosen because its objectives conflict by construction—coverage and conciseness cannot both be maximised—and they do: their game gradients correlate 0.203 against a mean of 0.399 across all pairs, where the UltraFeedback panel averages 0.616. That conflict separates the rules here in a way it does not there: each rule now buys its favoured objective at a visible cost in another, and no arm is uniformly best. The SafeRLHF panel is not yet built and is omitted rather than left blank.

	TL;DR summarisation, four objectives					
method	coverage s	faithful s	concise s	helpful s	min s	avg s
NBPO (ours)	0.396	0.157	−0.001	0.357	−0.001	**0.227**
PROSPER/MaxEntBW	0.410	0.063	−0.137	0.351	−0.137	0.171
Utilitarian	**0.442**	0.144	−0.092	**0.377**	−0.092	0.218
Surplus maxmin	0.162	**0.181**	**0.036**	0.132	**0.036**	0.128

These tables serve different claims. Table 3 tests whether the learned compromise approaches the intended bargaining outcome under genuine objective conflict. Table 2 checks that gains are not caused by verbosity or broad capability regression. Table 8 tests whether the result transfers to safety and summarization rather than depending on one prompt distribution.

Table 9: Detailed finite-β HelpSteer2 results from the available matched run, on the population scale of (2). "IR" is the number of runs with positive improvement on every objective.

method	helpful s	correct s	coherent s	concise s	min s	F	IR
NBPO	0.197	0.134	0.104	0.121	0.104	−8.011	1/1
Utilitarian	0.199	0.132	0.109	0.125	0.109	−7.931	1/1
Absolute maxmin	0.221	0.164	0.129	0.198	0.129	−6.978	1/1
Surplus maxmin	0.233	0.159	0.133	0.139	0.133	−7.286	1/1
Fixed-reference precursor	0.223	0.154	0.134	0.154	0.134	−7.251	1/1
Uniform-INPO	0.170	0.099	0.080	0.079	0.079	−9.156	1/1

Remaining reporting checks. A final multi-seed evaluation should document the payoff normalization, plot each run's theoretical minimum surplus, Nash welfare, and dual residual across constrained stages, mark rejected steps, test opponent-pool sensitivity, audit generic or excessive-refusal behavior, and report both game values and conventional fixed-reference win rates.

Table 10: Detailed finite-β SAFERLHF results from the available matched run, on the population scale of (2). "IR" is the number of runs with positive improvement on every objective.

method	helpful s	harmless s	min s	F	IR
NBPO	0.500	0.534	0.500	-1.319	1/1
Utilitarian	0.510	0.527	0.510	-1.314	1/1
Absolute maxmin	0.496	0.536	0.496	-1.325	1/1
Surplus maxmin	0.497	0.531	0.497	-1.332	1/1
Fixed-reference precursor	0.520	0.535	0.520	-1.278	1/1
Uniform-INPO	0.495	0.535	0.495	-1.328	1/1

Table 11: Temperature and certification ablation on the population scale of (2). "Gap" is the reported difference from the fixed-reference precursor. It should not be compared numerically with Theorem 4 until the reporting-scale normalization is documented.

β	min s	F	reported gap	opp. entropy	accepted stages	ref. win
∞	0.520	-1.278	0	n/a	2/2	0.837
1.0	0.487	-1.369	-0.034	0.988	2/2	0.803
0.25	0.500	-1.319	-0.020	0.877	2/2	0.817
0.05	0.475	-1.385	-0.046	0.592	2/2	0.802
0 (finite game only)	0.569	-1.078	-0.019	0	n/a	n/a

Table 12: Raw-label scaling intervention. The coupled condition scales both A_k and β_k by λ and should preserve the population target; the fixed-temperature condition intentionally changes the comparator neighborhood.

condition	λ	target cosine	policy shift	min $\widetilde{s}$	opp. entropy
coupled (A_k, β_k)	0.6	1.000000	0.195	0.600	0.877
coupled (A_k, β_k)	0.3	1.000000	0.471	0.300	0.877
fixed β_k	0.6	0.993629	0.273	0.751	0.941
fixed β_k	0.3	0.972431	0.476	0.556	0.983

E ULTRAFEEDBACK-SCALE MATCHED PROJECTION STUDY

This study instantiates the matched-comparison protocol of Appendix D on 7,000 ULTRAFEEDBACK prompts with four objectives (instruction following, truthfulness, honesty, helpfulness), a pinned `Qwen3-32B` guided-choice judge queried in both presentation orders, four candidate responses and four reference anchors per prompt (896,000 judgments per side), and a 6,000/1,000 train/held-out prompt split. All aggregation rules share one fixed parent policy, one frozen pair set, and one optimizer budget.

Target reliability across independent candidate halves. Table 13 reports the split-half reproducibility of each rule's centered per-response target: the target is computed twice from disjoint two-candidate halves, and the two versions are correlated over all held-out responses with a paired 2,000-resample prompt bootstrap. Every objective has strictly positive policy-minus-reference surplus in each half. The NBPO target is significantly more reproducible than both maxmin targets and statistically indistinguishable in practical terms from the utilitarian target, consistent with the near-uniform inverse-surplus weights that a balanced panel induces.

Table 13: Split-half reliability of the centered selection targets on 7,000 ULTRAFEEDBACK prompts. "argmax" is the fraction of prompts whose top-scored response agrees between halves. Δ is NBPO minus the row's rule (paired bootstrap 95% interval).

selection rule	split-half Pearson [95% CI]	argmax	Δ vs. NBPO [95% CI]
NBPO	0.884 [0.879, 0.888]	0.787	—
Utilitarian	0.894 [0.890, 0.897]	0.798	-0.010 [-0.012, -0.009]
Absolute maxmin	0.869 [0.864, 0.875]	0.932	$+0.014$ [$+0.008$, $+0.020$]
Surplus maxmin	0.639 [0.629, 0.649]	0.617	$+0.244$ [$+0.236$, $+0.253$]

Held-out realization of the regression target. Table 14 measures whether the trained network reproduces its prescribed mirror target on held-out prompts: the pairwise policy log-ratio is correlated against the frozen centered target on 148 held-out prompt groups after training on 1,340 groups for exactly four passes. Same-split memorization unit tests (32 groups, 50 passes) exceed Pearson 0.98 for every configuration shown, so optimizer and capacity are not the binding constraint; the target itself has split-half reliability above 0.88, so label noise is not the ceiling either. Across the logistic and squared losses, LoRA ranks 64 and 256, and a full-precision forward pass, held-out realization plateaus near 0.36, far below the 0.80 preregistered for opening reward-based selection. We are not aware of prior preference-optimization work that reports this quantity; the plateau motivates the pair-oriented projection used in Table 3.

Table 14: Held-out realization of the prescribed regression target (ULTRAFEEDBACK, 1,340 train / 148 held-out prompt groups, four passes, best learning rate per grid, fp32 forward). "KL" is the median per-prompt KL from the parent.

projection	adapter	train Pearson	held-out Pearson	held-out KL
Logistic (Bradley–Terry)	LoRA-64	0.954	0.366	0.018
Squared (23)	LoRA-64	0.953	0.366	0.018
Squared (23)	LoRA-256	0.928	0.363	0.018

Rescaling test: construction and caveats. Theorem 2 states that the NBPO solution is unchanged when each objective's utility is rescaled independently. Table 4 tests this on the frozen pool without retraining or relabelling: each objective's payoff and temperature are multiplied by its own factor λ_k, which leaves the underlying preferences untouched, and every rule's selection is recomputed. We report the fraction of the $6{,}000 \times 6$ response pairs whose preferred element changes. NBPO does not move a single pair under any of the three rescalings, matching the theorem exactly. Every competing rule does: the utilitarian average and PROSPER's per-prompt worst objective both compare raw values across objectives, and the two maxmin rules move most of all. The zeros in the maxmin rows are coincidental rather than structural—they occur only when a rescaling happens to leave the arg-minimising objective unchanged—whereas the NBPO zeros hold for every rescaling. The prompt-wise NBPO variant is close to invariant but not exactly so, because a prompt whose estimated surplus is nonpositive needs a numerical floor and the floor does not commute with rescaling; the same floor is what makes the panel-level rule deviate when it binds. Two of the added baselines land on the utilitarian row exactly, and for reasons worth stating. Uniform-INPO differs from the utilitarian rule in its training loss, not in how it combines objectives, so it inherits that rule's scale sensitivity unchanged. MOPO's log-barrier multipliers $\lambda_k = \sigma / \max(G_k - b_k, \sigma^2)$ never bind at these surplus magnitudes—$G_k - b_k \approx 0.05$ against $\sigma^2 = 1$—so its aggregation also reduces to the uniform sum. The HT-MNPO corners are exactly invariant, but only because a rule that reads a single objective cannot be disturbed by rescaling that objective: they buy invariance by ignoring the other objectives, and Table 3 shows what that costs them on individual rationality.

The rescaling vectors, in the objective order (helpfulness, honesty, instruction following, truthfulness), are $\lambda^{(A)} = (0.2, 0.6, 1.0, 1.7)$, $\lambda^{(B)} = (3.0, 0.5, 1.0, 0.25)$ and $\lambda^{(C)} = (1.0, 1.0, 1.0, 5.0)$.

Matched game-value comparison. Table 3 in Appendix D.1 evaluates all arms on the 1,000 held-out prompts under the frozen evaluator, anchors, and decoding shared with the parent run. "pair-projection" arms orient each frozen response pair by the rule's selector score and train with a standard length-normalized preference loss; "exact regression" arms regress the pairwise log-ratio difference onto the rule's signed score difference with the manuscript loss (23). Both use the identical 1,200-step budget. The completed rows stress-test the certificate requirement of Algorithm 1. Run open-loop at four times the manuscript's optimizer budget, the squared-loss update drives every rule to the same collapsed policy: mean response length grows from 1,399 to about 2,700 characters, the anchored win rate falls to 0.07–0.09, and every held-out surplus is strongly negative. Restoring the manuscript's own 300-step budget removes the collapse—response lengths stay within 1.2–$1.3\times$ the parent and generations remain fluent—and the aggregation rules then separate measurably. NBPO dominates the utilitarian rule on every objective, on the minimum surplus, and on the anchored win rate, which is the comparison the inverse-surplus weights are designed to affect; both maxmin rules place ahead of NBPO on this panel, with surplus maxmin closest to the fallback. Critically, no trained arm at either budget attains joint positive surplus, so the constrained algorithm would reject every candidate. Since the same collapse appears under every rule, it is attributable to the uncertified projection step rather than to the aggregation rule, and it is consistent with the held-out realization plateau of Table 14: when the fitted update does not track its prescribed target off-distribution, constrained stage acceptance is the mechanism that prevents drift. We therefore report these runs as a negative result about uncertified projection, not as evidence for or against the bargaining rule.

F Temperature and Step Size on SafeRLHF

SafeRLHF is the panel where NBPO places worst, so it is where the free parameters of the game construction deserve the most scrutiny. Every panel in this paper fixes $\beta = 0.25$, and that value was never justified against alternatives. This appendix sweeps it, and because raising it helps NBPO it applies the same temperature to the rule NBPO trails, from the same pool and the same re-solved dual: reporting our method at its best temperature against a baseline pinned at another would be the asymmetry this paper objects to for the step size.

Table 15 gives the result. Raising β from 0.25 to 0.5 helps NBPO a great deal, $+0.078$ on the minimum surplus, and helps absolute maxmin only $+0.010$, which narrows the gap between them from 0.086 to 0.017. It does not reverse it, and the trend does not continue: NBPO peaks at 0.5 and falls back at 1.0 and 2.0, so the rise across the first three temperatures was not a trend to extrapolate. Table 17 puts that residual gap in context: on this panel the top three arms are mutually indistinguishable at the resolution of a hundred evaluation prompts, so a gap of 0.017 is below what this evaluation can resolve and no setting of either knob could have been credited with closing it. The step size was then re-swept at $\beta = 0.5$, since it had been selected at 0.25 and the two interact; the previously selected value remains the best for NBPO and likewise for absolute maxmin.

Taken together the two knobs are exhausted for both arms, along with the optimizer budget, which halves NBPO's minimum surplus when doubled. Absolute maxmin leads this panel at every setting we have tried, and Table 17 confirms that NBPO's deficit there is significant while the differences among the three leading arms are not. We report that rather than presenting the single narrowest configuration as a win. It is consistent with what separates this panel from the others: it carries two objectives against four, and the share of response pairs on which every objective agrees is 0.50 here against 0.42 on ULTRAFEEDBACK and 0.25 on TL;DR. Where objectives rarely disagree there is little for a bargaining rule to arbitrate, and the rule that simply protects the weakest raw game value does as well or better.

Table 15: Game temperature and proximal step size on SAFERLHF. Held-out surpluses on 100 prompts, population scale of (2). The dual is re-solved at each temperature because the surpluses it inverts change with it, and both arms are retrained at every temperature from the same pool. The step size is the target RMS of the regression; the lower block re-sweeps it at the temperature that suits NBPO best. Each arm's selected setting is bold.

	NBPO (ours)		Absolute maxmin	
setting	min s	avg s	min s	avg s
temperature, at each arm's selected step size				
$\beta = 0.1$	0.1805	0.2176	—	—
$\beta = 0.25$	0.2022	0.2191	0.2878	0.2982
$\beta = 0.5$	$\mathbf{0.2806}$	$\mathbf{0.2851}$	0.2973	0.3009
$\beta = 1.0$	0.2209	0.2391	$\mathbf{0.2995}$	$\mathbf{0.3045}$
$\beta = 2.0$	0.2458	0.2606	0.2922	0.3025
step size, at $\beta = 0.5$				
target RMS 1.29	-0.0261	-0.0205	0.2739	0.2826
target RMS 3.44	0.2675	0.2680	—	—
selected RMS	$\mathbf{0.2806}$	$\mathbf{0.2851}$	$\mathbf{0.2973}$	$\mathbf{0.3009}$
target RMS 21.5	-0.0875	-0.0312	—	—

G A SECOND JUDGE

Every surplus in this paper is scored by the same prompted `Qwen3-32B` oracle that supplied the training preferences. Only the benchmark table uses a judge disjoint from training. That leaves the central claim—that bargaining raises the weakest objective—untested against a second evaluator, and a reviewer is entitled to ask whether it is a property of the method or of the oracle. Table 16 re-scores two panels with `Llama-3.3-70B-Instruct`, which supplied none of the training preferences. Decoding uses the same seed, so the responses are the ones already scored and only the judgment changes.

The two panels answer differently, and the difference is itself the finding. On ULTRAFEEDBACK/`Qwen3-8B` the ordering is preserved and the numbers barely move: NBPO leads under both judges, and its margin over the runner-up is *larger* under the disjoint judge ($+0.019$ against $+0.014$), so the training oracle was not flattering it. On TL;DR the panel does not transfer. The second judge is far harsher on conciseness—surplus maxmin falls from $+0.036$ to -0.055, the utilitarian rule from -0.092 to -0.362—so no arm attains individual rationality under it, and the ordering reshuffles: NBPO sits third under the training oracle and fourth under the disjoint one. We therefore state the TL;DR panel's individual-rationality verdicts as judge-dependent, and rest the cross-judge claim on ULTRAFEEDBACK, where it holds.

One caveat on the TL;DR NBPO row. That arm's checkpoint was lost and had to be retrained from the same pool, dual and configuration; its precomputed pair set had also been deleted, so the pairs were rebuilt, and the restored arm reaches -0.038 on the training oracle against the original's -0.006. The row above uses the restored arm under both judges, so the comparison between judges is internally consistent, but it is not the same weights that produced the main table's TL;DR entry. The published checkpoint is the restored one.

H BOOTSTRAP INTERVALS ON THE HELD-OUT SURPLUSES

The surpluses in Table 3 are averages over one hundred held-out prompts, and until now they carried no interval. That mattered: the benchmark table's gaps were already known to sit inside a per-arm

Table 16: Minimum held-out surplus under the training oracle and under a judge disjoint from training, population scale of (2), same responses in both columns. Arms are ordered by the disjoint judge. "Δ" is the change induced by swapping the judge.

panel	method	`Qwen3-32B`	`Llama-3.3-70B`	Δ
ULTRAFEEDBACK `Qwen3-8B`	NBPO (ours)	**0.036**	**0.033**	−0.002
	Absolute maxmin	0.009	0.015	+0.006
	Utilitarian	0.021	0.012	−0.010
	Surplus maxmin	0.008	0.010	+0.002
	MOPO	−0.009	−0.001	+0.008
	Uniform-INPO	−0.010	−0.007	+0.003
	PROSPER/MaxEntBW	−0.004	−0.011	−0.007
TL;DR	Surplus maxmin	**0.036**	**−0.055**	−0.091
	MOPO	−0.001	−0.083	−0.082
	Uniform-INPO	−0.104	−0.117	−0.014
	NBPO (ours)	−0.038	−0.311	−0.273
	Utilitarian	−0.092	−0.362	−0.270
	PROSPER/MaxEntBW	−0.137	−0.372	−0.235
	Absolute maxmin	−0.159	−0.402	−0.243

seed standard deviation of 0.014, and without intervals on the surpluses there was no way to tell which of their orderings were real either. Table 17 resamples prompts, paired across arms so that a difference reflects the arms rather than which prompts were drawn, and recomputes the panel statistic inside each replicate.

Three of the paper's readings survive and one does not. On ULTRAFEEDBACK/`Qwen3-8B` NBPO has the largest minimum surplus and is significantly ahead of PROSPER, uniform-INPO and MOPO, but its margins over the utilitarian rule, absolute maxmin and surplus maxmin all straddle zero: it leads that panel, but not distinguishably so. On TL;DR and SAFERLHF its deficits against the panel leader are significant, -0.038 and -0.086, so those losses are real rather than noise. On SAFERLHF the top three arms are mutually indistinguishable, which also settles a question raised by the sweeps of Appendix F: the 0.008 gap those sweeps were narrowing was smaller than the resolution of the evaluation, so no setting of the temperature or the step size could have been credited with closing it.

I HOW FAR THE SCALE INVARIANCE SURVIVES THE PIPELINE

Table 4 rescales each objective on a frozen pool and recomputes which response each rule prefers. That is a statement about the selection rule, and it is the statement Theorem 2 makes. It is not the statement a reader is likely to take from it, which is that a policy trained under rescaled objectives is the same policy. Those differ by everything the pipeline puts between the rule and the weights: a dual solved numerically rather than analytically, a finite pair sample, and a stochastic optimizer. This appendix carries the same rescaling $\lambda_k^{(A)} = (0.2, 0.6, 1.0, 1.7)$ all the way through training and reports how much invariance is left at each stage.

The apparatus is the published builder and dual solver with one addition, a `--scales` argument that multiplies each objective's game value before aggregation. On the identity rescaling the patched code reproduces the paper's dual to a maximum absolute difference of 0.000000, so the comparison below is between rescalings and not between implementations. Four arms train on the 7,000-prompt ULTRAFEEDBACK pool for 300 steps from $\pi_0 = \mu$: the bargaining rule and the utilitarian rule, each on the identity pool and on the rescaled pool. The utilitarian arms are the control. Its aggregation has no cancellation to perform, so if its two arms also came out identical the experiment would be measuring the optimizer's insensitivity rather than the rule's invariance.

The dual does absorb the rescaling as the theory says. Its multipliers move from $(16.599, 14.455, 15.911, 18.038)$ to $(82.665, 24.073, 15.899, 10.602)$, against the predicted λ_k/a_k of $(82.993, 24.092, 15.911, 10.611)$. What Table 18 shows is that the invariance this buys is exact at the top of the pipeline and progressively less visible below it.

Table 17: Held-out surplus with 95% bootstrap intervals, 2000 replicates resampling the one hundred evaluation prompts, paired across arms. The right column is the difference in minimum surplus against that panel's leading arm, with its own paired interval; "sig." marks intervals excluding zero.

panel	method	min s [95% CI]	vs. leader
TL;DR	Surplus maxmin	0.036 [0.008, 0.065]	leader
	NBPO (ours)	−0.001 [−0.028, 0.027]	−0.038 sig.
	MOPO	−0.001 [−0.026, 0.024]	−0.038 sig.
	Utilitarian	−0.092 [−0.124, −0.059]	−0.128 sig.
	Uniform-INPO	−0.104 [−0.151, −0.074]	−0.140 sig.
	PROSPER/MaxEntBW	−0.137 [−0.169, −0.104]	−0.173 sig.
	Absolute maxmin	−0.159 [−0.191, −0.125]	−0.195 sig.
SafeRLHF	Absolute maxmin	0.288 [0.251, 0.324]	leader
	Surplus maxmin	0.287 [0.245, 0.323]	−0.001
	Utilitarian	0.276 [0.237, 0.314]	−0.011
	PROSPER/MaxEntBW	0.232 [0.194, 0.271]	−0.056 sig.
	NBPO (ours)	0.202 [0.162, 0.244]	−0.086 sig.
	MOPO	0.191 [0.156, 0.228]	−0.096 sig.
	Uniform-INPO	−0.154 [−0.194, −0.118]	−0.442 sig.
UltraFeedback `Qwen3-8B`	NBPO (ours)	0.036 [0.006, 0.064]	leader
	Utilitarian	0.021 [−0.010, 0.037]	−0.015
	Absolute maxmin	0.009 [−0.017, 0.032]	−0.027
	Surplus maxmin	0.008 [−0.019, 0.026]	−0.028
	PROSPER/MaxEntBW	−0.004 [−0.035, 0.025]	−0.040 sig.
	MOPO	−0.009 [−0.031, 0.007]	−0.045 sig.
	Uniform-INPO	−0.010 [−0.035, 0.004]	−0.046 sig.

Table 18: Invariance of NBPO under $\lambda^{(A)} = (0.2, 0.6, 1.0, 1.7)$, measured at three points in the pipeline, with the utilitarian rule as the control. Rows are ordered by how much machinery separates the measurement from Theorem 2. "Sign flip" is the percentage of the 37,800 training pairs whose regression target changes sign under rescaling; "max $|\Delta|$" is the largest change in any one objective's held-out surplus between the two trained arms, and "Δ min s" the change in the minimum. Held-out quantities are on 100 prompts under the pinned `Qwen3-32B` oracle, the protocol of Table 3, whose paired bootstrap resolution on a minimum surplus is approximately ± 0.03—wider than either rule's movement in that block. The generated-text block decodes greedily at temperature 0 and asks how often the rescaled arm reproduces its unrescaled twin token for token, alongside the floor obtained by retraining the unrescaled arm on identical targets with a different seed. Its rows are not bolded because the informative contrast there is between an effect and its own floor within a rule, not between the two rules.

level	quantity	NBPO	Utilitarian
Selection rule, dual set analytically to λ_k/a_k			
target correlation	Pearson r	0.999953	—
target sign flip	% of pairs	**0.217**	—
Estimated rule, dual re-solved numerically on the rescaled pool			
target correlation	Pearson r	**0.988626**	0.972373
target sign flip	% of pairs	**3.447**	5.624
Trained policy, held-out surplus after 300 steps			
objective-wise drift	max $\|\Delta\|$	**0.0154**	0.0260
minimum surplus	Δ min s	+0.0048	**−0.0004**
Generated text, greedy decoding on 500 shared prompts			
rescaled vs. identity	exact match %	7.0	4.4
seed replicate floor	exact match %	2.4	0.4
rescaled vs. identity	prefix agree %	23.9	20.5
seed replicate floor	prefix agree %	14.6	6.8

Three readings follow, and only the first is a confirmation. At the level the theorem speaks about, the invariance is essentially exact: fixing the dual at its analytic value leaves 99.8% of the regression targets with their sign, and the residual is the floor of the arithmetic. One stage down, once the dual is re-solved numerically from $\lambda = \mathbf{1}$ on the rescaled pool as a real run must, the exactness is gone and what remains is a margin over the control—3.4% of targets flip against the utilitarian rule's 5.6%. The invariance is therefore a property of the solution, and only inherited by a run to the extent that the run finds it.

At the level of trained policies the test does not separate the two rules. NBPO's largest per-objective movement, 0.0154, is smaller than the utilitarian rule's 0.0260, which is the predicted direction, but its minimum surplus moves $+0.0048$ where the control moves -0.0004, which is the opposite one. Both magnitudes sit well inside the ± 0.03 bootstrap resolution of this evaluation (Appendix H), so a hundred held-out prompts cannot resolve a difference of this size in either direction.

Greedy decoding is the sharper instrument, since two policies that are the same function emit the same tokens and no judge stands in the way, and it is worth reporting what it does and does not establish. The rescaled NBPO arm reproduces its unrescaled twin exactly on 7.0% of prompts. Read alone that number invites the conclusion that the invariance is absent, and that reading is wrong: two NBPO runs that differ *only* in random seed, on identical targets, agree on 2.4%. Rescaling every objective therefore perturbs the trained policy *less* than rerunning training does, and the same holds for the control (4.4% against a 0.4% floor). Both rules are thus invariant to within training stochasticity, and this measurement cannot distinguish them either—if anything the control sits further above its own floor than NBPO does above its.

What this bounds is not the theorem but the claim a frozen-pool table can support. Table 4 establishes that the selection rule is invariant, and that result is exact and reproducible. We cannot presently show that a policy trained through the rule inherits the invariance to any accuracy finer than the noise of retraining, because at a 300-step budget on an 8B model that noise is large: a seed change alone leaves only 2.4% of greedy decodes intact. Separating the two would require either a converged optimizer or many seeds per arm, and we report the negative rather than presenting the frozen-pool result as though it settled the trained-policy question.

Two further cautions about this panel. The utilitarian arms attain the larger surpluses here (minimum $+0.027$ against -0.008), which is not a comparison the paper makes elsewhere and should not be read as one: these are 300-step arms trained on the 7,000-prompt pool and evaluated on 100 prompts, a budget and panel combination used nowhere else, and the arms exist to be compared with their own rescaled twins rather than with each other. And the whole construction rescales the objectives inside the pipeline, where the payoffs are already estimates; a genuine change of units upstream of the judge would also change the labels, which is the separate question treated in Appendix A.4.

J OBJECTIVE-WISE SURPLUSES

Table 3 reports each panel's minimum and average surplus. This table gives the objective-by-objective breakdown those two statistics summarise, on the same evaluations and the same population scale, and adds the `Qwen3-8B` replication of the ULTRAFEEDBACK panel that the main-text table omits so that all of its rows share one backbone. Objective columns follow each panel's own header; SAFERLHF has two objectives, so its last two columns are empty. Reading down a panel shows which objective each rule sacrifices: the utilitarian and maxmin rules buy helpfulness on ULTRAFEEDBACK at the cost of the other three, PROSPER and the utilitarian rule drive conciseness clearly negative on TL;DR, and the single-objective corners lose whichever objectives they do not optimise.

K FIXED-REFERENCE PRECURSOR EXPERIMENTS

The tables below preserve the numerical results from the preceding NBPO draft. These runs use the anchor margin $a_k(\pi) = P_k(\pi \succ \mu) - 1/2$, add $-0.05\mathrm{KL}(\pi\|\mu)$ to the welfare objective, initialize directly at μ, and use clipped inverse-surplus weights. By Theorem 4, they concern the $\beta = \infty$ endpoint, not finite-temperature NBPO. They are retained as an audit trail and must not be cited as evidence for the regularized-opponent objective.

Table 19: Objective-wise held-out surplus over the reference for every arm and every panel, population scale of (2). Within a panel each arm starts at $\pi_0 = \mu$, trains on the identical response pool with the identical optimizer budget, and differs only in the weight vector over objectives. Across panels the magnitudes are not comparable: ULTRAFEEDBACK on `Llama` is a β-regularised game-value difference on a 657-prompt jointly complete panel, while the other three are anchored win-rate margins $P_k - \frac{1}{2}$ on 100 held-out prompts against four anchor samples. Bootstrap intervals for the latter three are in Table 17.

method	obj. 1	obj. 2	obj. 3	obj. 4	min s	avg s
ULTRAFEEDBACK on `Llama-3.1-8B-Instruct`: instruction following, truthfulness, honesty, helpfulness						
NBPO (ours)	0.039	0.049	0.055	0.109	0.039	0.063
PROSPER/MaxEntBW	−0.035	−0.051	−0.046	0.068	−0.051	−0.016
Utilitarian	−0.064	−0.077	−0.070	0.027	−0.077	−0.046
Absolute maxmin	−0.043	−0.062	−0.051	0.059	−0.062	−0.024
Surplus maxmin	−0.014	−0.000	0.020	0.108	−0.014	0.028
Uniform-INPO	−0.008	0.022	0.020	0.079	−0.008	0.028
MOPO	0.031	0.026	0.022	0.032	0.022	0.028
HT-MNPO (helpfulness)	−0.057	−0.073	−0.067	0.050	−0.073	−0.037
HT-MNPO (truthfulness)	−0.050	−0.049	−0.041	0.023	−0.050	−0.029
HT-MNPO (honesty)	0.022	0.027	0.025	0.075	0.022	0.037
HT-MNPO (instr. following)	0.008	0.026	0.032	0.065	0.008	0.033
ULTRAFEEDBACK on `Qwen3-8B`: helpfulness, truthfulness, honesty, instruction following						
NBPO (ours)	0.144	0.036	0.085	0.061	0.036	0.082
PROSPER/MaxEntBW	0.079	−0.003	0.032	0.034	−0.003	0.035
Utilitarian	0.076	0.024	0.032	0.023	0.023	0.039
Absolute maxmin	0.082	0.010	0.036	0.023	0.010	0.038
Surplus maxmin	0.075	0.008	0.029	0.017	0.008	0.032
Uniform-INPO	−0.010	−0.002	−0.002	−0.006	−0.010	−0.005
MOPO	−0.003	−0.001	−0.009	0.002	−0.009	−0.003
HT-MNPO (helpfulness)	−0.004	−0.002	−0.002	0.004	−0.004	−0.001
HT-MNPO (truthfulness)	0.034	0.002	0.033	0.028	0.002	0.024
HT-MNPO (honesty)	0.001	0.010	0.003	−0.005	−0.005	0.002
HT-MNPO (instruction following)	−0.008	0.009	0.001	0.004	−0.008	0.001
TL;DR on `Llama-3.1-8B-Instruct`: coverage, faithfulness, conciseness, helpfulness						
NBPO (ours)	0.396	0.157	−0.001	0.357	−0.001	0.227
PROSPER/MaxEntBW	0.410	0.062	−0.137	0.350	−0.137	0.171
Utilitarian	0.442	0.144	−0.092	0.377	−0.092	0.218
Absolute maxmin	0.426	0.010	−0.159	0.363	−0.159	0.160
Surplus maxmin	0.162	0.180	0.036	0.132	0.036	0.128
Uniform-INPO	−0.102	−0.104	−0.022	−0.099	−0.104	−0.082
MOPO	0.122	0.018	−0.001	0.103	−0.001	0.060
HT-MNPO (coverage)	0.101	−0.020	−0.015	0.090	−0.020	0.039
HT-MNPO (faithfulness)	0.147	−0.041	−0.052	0.119	−0.052	0.043
HT-MNPO (conciseness)	0.252	−0.075	−0.158	0.212	−0.158	0.058
HT-MNPO (helpfulness)	0.073	−0.050	−0.070	0.096	−0.070	0.013
SAFERLHF on `Llama-3.1-8B-Instruct`: helpfulness, harmlessness						
NBPO (ours)	0.236	0.202	—	—	0.202	0.219
PROSPER/MaxEntBW	0.303	0.232	—	—	0.232	0.268
Utilitarian	0.314	0.276	—	—	0.276	0.295
Absolute maxmin	0.309	0.288	—	—	0.288	0.298
Surplus maxmin	0.300	0.287	—	—	0.287	0.294
Uniform-INPO	−0.154	−0.110	—	—	−0.154	−0.132
MOPO	0.203	0.191	—	—	0.191	0.197
HT-MNPO (helpfulness)	−0.048	−0.098	—	—	−0.098	−0.073
HT-MNPO (harmlessness)	−0.047	−0.100	—	—	−0.100	−0.074

Population optimizer audit. The full and minimal exact-gradient updates were identical, and absorbing the sum of inverse-surplus weights into the step size changed the policy by at most $2.9e-15$ in ℓ_1. A stage-level acceptance test reduced regret under frozen gradient noise by 0.0372, whereas raw inverse weights at the same nominal step reduced the strict-IR rate by 4.0%. The scale-covariance and Pareto residuals were below $2.4e - 15$. These archived checks support preserving the exact coefficient scale and retaining stage acceptance, but they do not evaluate the new constrained proximal solver.

The simultaneous Hoeffding lower bound made no false certifications in this study, compared with 1.09% for the plug-in rule at $n = 128$, but its power was only 9.0% at $n = 8192$. We therefore use confidence bounds as external certificates rather than as an optimization ingredient.

Table 20: Four-objective HELPSTEER2 fixed-reference precursor results with a prompted pairwise oracle. Entries report win rates against the reference, averaged over `Phi-4` and `Llama-3.3-70B`, neither of which supplied training preferences. The upper block reports four-stage runs, while the lower block reports single-stage baselines for context. Results are from one training seed.

rule	helpful	correct	coherent	concise	Avg	Worst
four stages each						
NBPO-KL	0.660	0.642	0.555	0.471	0.582	0.471
Utilitarian	0.625	0.597	0.530	0.426	0.545	0.426
MaxMin	0.463	0.424	0.382	0.369	0.410	0.369
single stage						
Utilitarian, 1 stage	0.599	0.583	0.559	0.416	0.539	0.416
MaxMin, 1 stage	0.604	0.587	0.559	0.420	0.543	0.420
HT-MNPO-helpful	0.654	0.618	0.593	0.324	0.547	0.324
HT-MNPO-correct	0.629	0.617	0.574	0.381	0.550	0.381
HT-MNPO-coherent	0.631	0.596	0.579	0.351	0.539	0.351
HT-MNPO-concise	0.422	0.470	0.448	0.722	0.515	0.422
INPO	0.616	0.604	0.560	0.425	0.551	0.425
SimPO	0.618	0.610	0.560	0.429	0.554	0.429
IPO	0.609	0.589	0.553	0.424	0.544	0.424
DPO	0.603	0.574	0.561	0.417	0.539	0.417
Base (μ)	0.500	0.500	0.500	0.500	0.500	0.500

Table 21: Three training seeds of the two four-stage fixed-reference rules on HELPSTEER2. Scored by `Phi-4` on 500 held-out prompts. The $\pm$ figures are standard deviations over training seeds. The ordering does not persist across seeds.

rule	seed 42	seed 43	seed 44	mean
worst objective				
NBPO-KL	0.504	0.398	0.328	$0.410_{\pm.088}$
Utilitarian	0.461	0.490	0.318	$0.423_{\pm.092}$
average objective				
NBPO-KL	0.593	0.449	0.385	$0.475_{\pm.106}$
Utilitarian	0.551	0.601	0.373	$0.508_{\pm.120}$

Interpretation. The precursor runs provide no reliable finite-β comparison. The three-seed HELPSTEER2 ordering changes sign, and repeated fixed-reference updates can leave the positive-surplus region, especially when the effective step scale is not controlled. These observations motivate matched seed-level evaluation, constrained proximal updates, and stage-wise certification, but they are not performance claims for NBPO.

Table 22: SAFERLHF fixed-reference precursor results. Entries are win rates against `Llama-3.1-8B-Instruct` on 500 held-out prompts, averaged over `Phi-4` and `Llama-3.3-70B`. The subscript on NBPO-KL is the spread over three training seeds.

rule	helpful	harmless	Avg	Worst
NBPO-KL	0.807	0.845	0.826	$0.807_{\pm.014}$
Utilitarian	0.816	0.845	0.830	0.816
MaxMin	0.806	0.853	0.830	0.806
HT-MNPO-helpful	0.803	0.784	0.794	0.784
HT-MNPO-harmless	0.769	0.853	0.811	0.769
INPO	0.808	0.816	0.812	0.808
SimPO	0.780	0.764	0.772	0.764
IPO	0.753	0.727	0.740	0.727
DPO	0.704	0.716	0.710	0.704
Base (μ)	0.500	0.500	0.500	0.500

Table 23: Fixed-reference target under symmetric coin-flip contamination of the helpfulness judge on HELPSTEER2. The exact proportionality is a property of the $\beta = \infty$ anchor limit. It is not the prediction for finite-temperature NBPO with fixed β.

rule	λ	ω helpful	ω concise	mean $\lvert t \rvert$	cosine	ratio spread
NBPO-KL	1.0	0.076	0.470	0.393	n/a	n/a
NBPO-KL	0.6	0.120	0.447	0.374	1.000000	8×10^{-15}
NBPO-KL	0.3	0.214	0.399	0.334	1.000000	5×10^{-15}
Utilitarian	1.0	0.250	0.250	0.403	n/a	n/a
Utilitarian	0.6	0.250	0.250	0.363	0.999105	1.28
Utilitarian	0.3	0.250	0.250	0.332	0.996738	2.24
MaxMin	any	0.000	1.000	0.411	1	0

Table 24: Fixed-reference ablation of weight normalization and pair selection on SAFERLHF. Entries are reference win rates on 500 held-out prompts. The sharp degradation without normalization motivates treating the normalized coefficient as an explicit effective step size.

			`Qwen3-32B`		`Phi-4`	
weights	pairs	stage	Avg	Worst	Avg	Worst
normalized	best-of-4	1	0.737	0.722	0.772	0.747
normalized	best-of-4	2	0.698	0.684	0.724	0.710
normalized	random	1	0.730	0.699	0.745	0.707
normalized	random	2	0.667	0.635	0.665	0.608
unnormalized	best-of-4	1	0.490	0.457	0.499	0.490
unnormalized	best-of-4	2	0.095	0.045	0.032	0.028
unnormalized	random	1	0.510	0.478	0.508	0.503
unnormalized	random	2	0.380	0.349	0.363	0.355

Table 25: Preregistered population-optimizer audit over 200 finite cyclic games. Regret differences are paired means with 95% bootstrap intervals.

comparison	endpoint	result
full vs. minimal	max policy ℓ_1	$0.0e + 00$
normalized vs. absorbed step	max policy ℓ_1	$2.9e - 15$
noisy stage gate	Nash-regret change	-0.0372 $[-0.0407, -0.0340]$
raw fixed step	strict-IR change	-0.040 $[-0.070, -0.015]$

L PROXIMAL STEP-SIZE SWEEP

Theorem 3 holds for arbitrary $\eta_t > 0$: the step size enters only through $1/\sum_t \eta_t$ in the rate, not as a constraint on validity. It is therefore an algorithm parameter rather than a tuned hyperparameter, and the value used in Table 2 is the one this sweep selects. All five settings share the dual solution, the pair set, the optimizer and the 300-step budget; only the coefficient multiplying the regression target changes.

Table 26: Held-out effect of the proximal step size on the ULTRAFEEDBACK panel (711 jointly complete prompts, population scale). "ratio" is the mean Arena-Hard generation length divided by the base policy's. Minimum surplus rises sevenfold to 20η and then falls, while the length ratio decreases monotonically to 20η; at 50η the regression target has RMS 21.5, the scale at which the archived open-loop arms collapsed at four times their budget, yet generations remain fluent.

step size	instr. s	truth. s	honest s	helpful s	min s	ratio
η	0.0086	0.0051	0.0207	0.1037	0.0051	1.03
3η	0.0324	0.0231	0.0366	**0.1189**	0.0231	1.02
8η	0.0267	0.0429	0.0519	0.1113	0.0267	1.02
20η	**0.0390**	0.0480	**0.0542**	0.1096	**0.0390**	**1.01**
50η	0.0095	**0.0547**	0.0505	0.1024	0.0095	1.03

What happens when every rule is given twice the budget. The step-size sweep varies η at a fixed 300-step budget. Table 27 varies the budget instead, doubling it to 600 steps for NBPO and for the two strongest baselines. Each arm is run from its own configuration with `max_steps` as the only field changed, so nothing distinguishes the doubled-budget arm from the published one except the budget. The rules do not respond alike. PROSPER and the utilitarian rule both collapse: every objective falls to roughly -0.09 and individual rationality is lost outright. NBPO at 8η keeps all four objectives positive and improves its average surplus, and only loses individual rationality at the largest step size, where the total displacement $\eta \times$ steps is twice that of any setting we report. The asymmetry is not explained by displacement alone: the baselines' targets are normalised to an RMS of 0.43 against NBPO's 3.44, so they move the policy *less* per step and still fail first. Their selectors are winner-take-all—PROSPER keeps only each prompt's worst objective and the utilitarian rule a single fixed average—and a longer run against such a target appears to drive the policy off the distribution the target was estimated on, whereas the bargaining weights keep every objective represented in the regression. We report the measurement and offer that reading as a hypothesis rather than a demonstration.

Table 27: Doubling the optimizer budget from 300 to 600 steps. Held-out surpluses on 100 prompts under the pinned `Qwen3-32B` oracle, population scale of (2). Each arm is trained from its own published configuration with `max_steps` as the only altered field; η is the proximal step size as a multiple of the base target RMS. "IR" marks whether all four objectives remain positive.

method	setting	instr.	truth.	honest	min s	IR
NBPO	20η, 300 steps	0.0370	**0.0236**	0.0318	**0.0236**	yes
NBPO	8η, 600 steps	0.0332	0.0232	**0.0434**	0.0232	yes
NBPO	20η, 600 steps	-0.0202	0.0006	-0.0036	-0.0202	no
PROSPER/MaxEntBW	600 steps	-0.1716	-0.1870	-0.1926	-0.1926	no
Utilitarian	600 steps	-0.1764	-0.1800	-0.1842	-0.1842	no

M JUDGE INSTANTIATION AND PREFERENCE LABELING

Training judges. Use the same prompted pairwise interface on both panels. For each objective, `Qwen3-32B` compares two responses under an attribute-specific rubric. Each pair is queried in both presentation orders, and the verdicts are swap-averaged. The fixed-reference precursor also contains deterministic Beaver reward and cost heads; those runs should not be conflated with the prompted-oracle main comparison.

Evaluation judges. Evaluation is separate from training and uses `Phi-4` and `Llama-3.3-70B`. Each evaluator returns `[[A]]`, `[[B]]`, or `[[TIE]]`, scored as 1, 0, or $1/2$, in both presentation orders. This makes the empirical evaluation oracle skew-symmetric after averaging.

Rubrics. Helpfulness asks which response better accomplishes the user's stated goal; correctness asks which is more accurate; coherence asks which is clearer and better organized; conciseness asks which conveys the needed content with less padding; and harmlessness asks which is safer. The judge is instructed to evaluate only the designated objective.

Prompt template. For concreteness, the helpfulness system message and pairwise user message are

```
System:  You compare two AI responses on HELPFULNESS ONLY:
which better helps the user accomplish their stated goal
(relevance, substance, actionable detail).  Ignore safety
entirely; a refusal with useful alternatives is more helpful
than a bare refusal.  End with exactly [[A]] or [[B]] or
[[TIE]].
User:  [User prompt] {prompt} [Response A] {A} [Response B]
{B} Verdict:
```

The other objectives use the same template with only the rubric changed, and each pair is queried again after swapping responses A and B.

From verdicts to regularized opponents. For a current policy π_t and a reference comparator z, estimate

$$\widehat{r}_{k,\pi_t}(z \mid x) = \frac{1}{L}\sum_{\ell=1}^{L}\left(B_k(y_\ell, z) - \tfrac{1}{2}\right), \qquad y_\ell \sim \pi_t(\cdot \mid x).$$

Reference responses are reweighted by $\exp\{-\widehat{r}_{k,\pi_t}(z \mid x)/\beta_k\}$ and normalized to sample $z_k \sim \widehat{\nu}^{\star}_{k,\pi_t}$. For a policy pair (y, y') sharing z_k, the binary difference in (21) enters the regression target directly. No scalar reward model or value function is fitted by NBPO.